

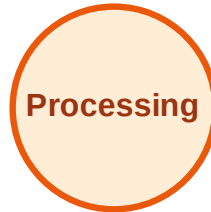
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



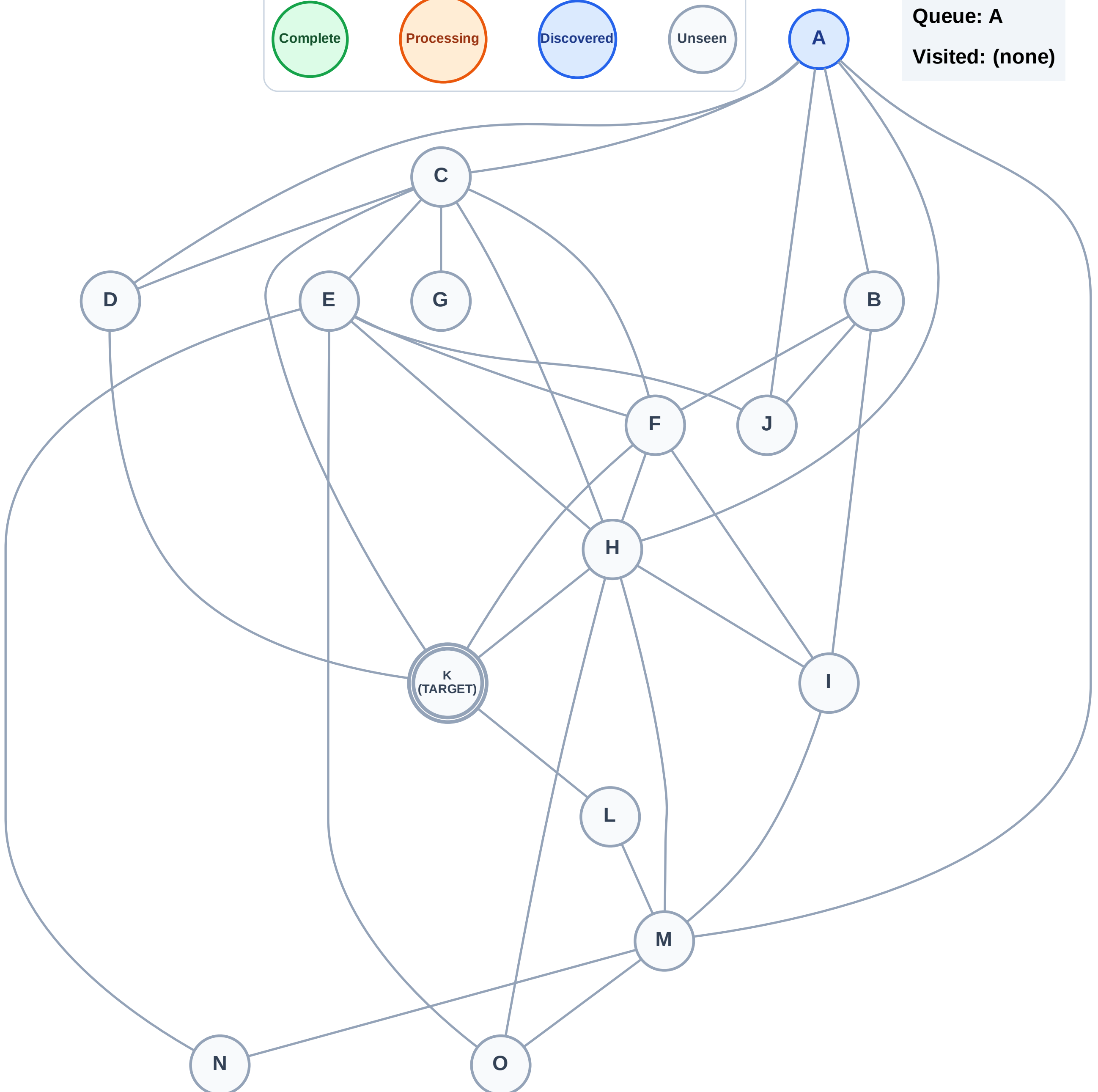
Discovered



Unseen

Queue: A

Visited: (none)



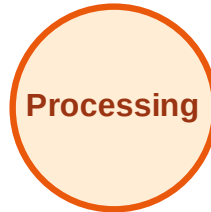
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



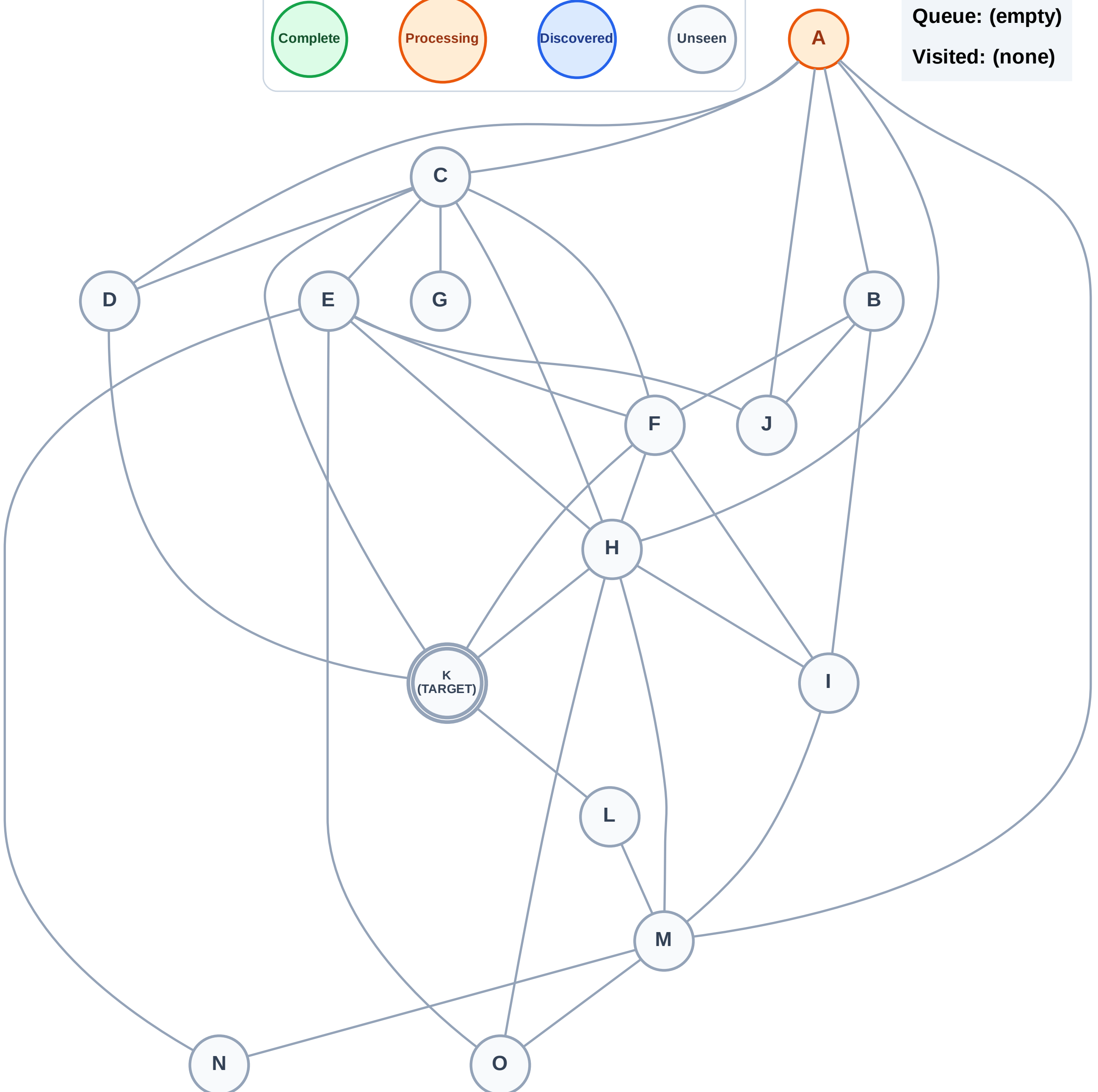
Discovered



Unseen

Queue: (empty)

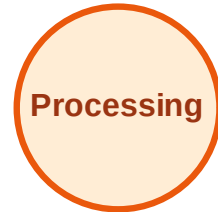
Visited: (none)



BFS Traversal

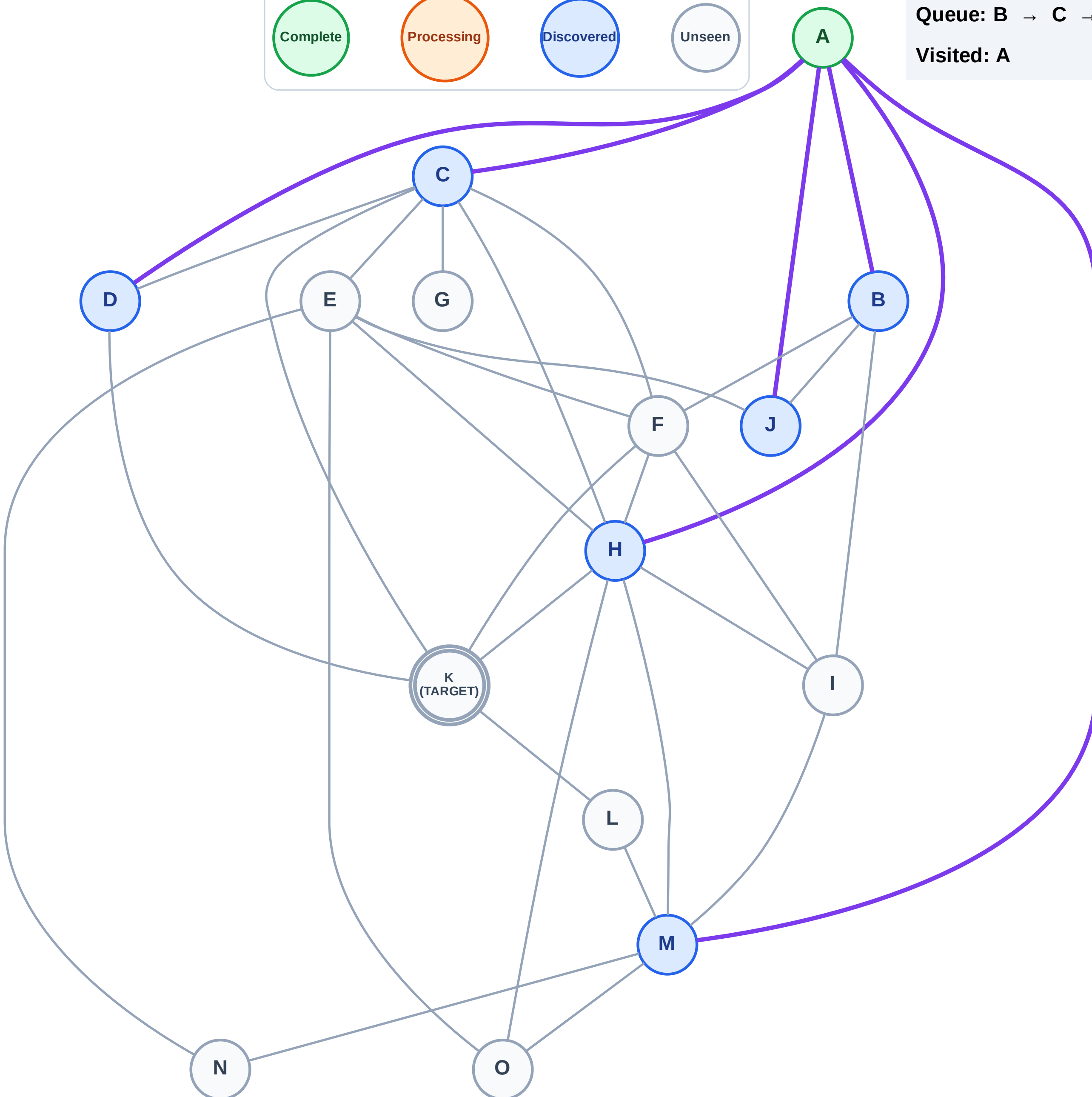
Step 3: Discover B, C, D, H, J, M from A

Legend



Queue: B → C → D → H → J → M

Visited: A



BFS Traversal

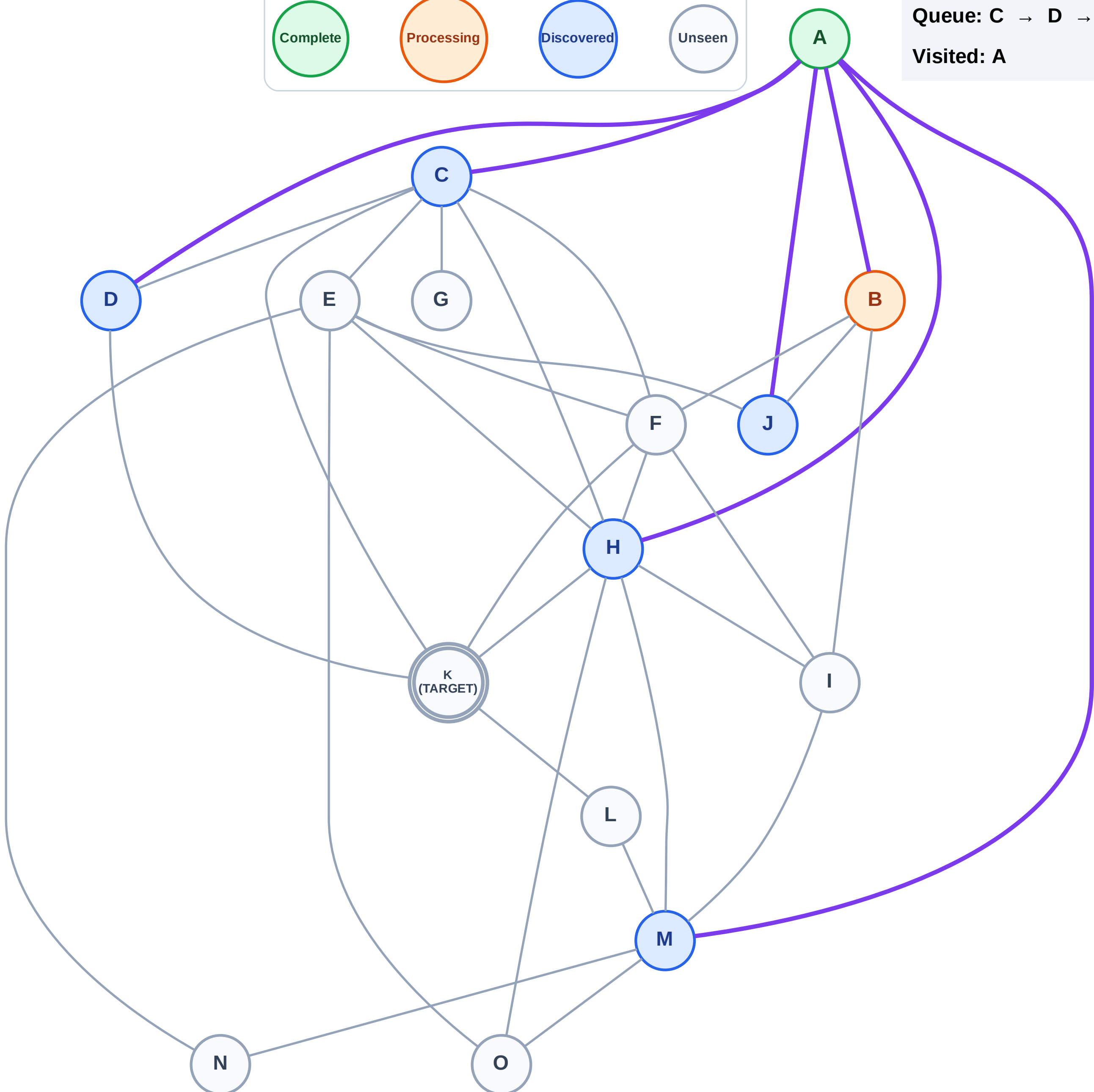
Step 4: Process node B

Legend



Queue: C → D → H → J → M

Visited: A



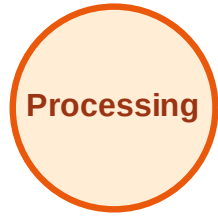
BFS Traversal

Step 5: Discover F, I from B

Legend



Complete



Processing



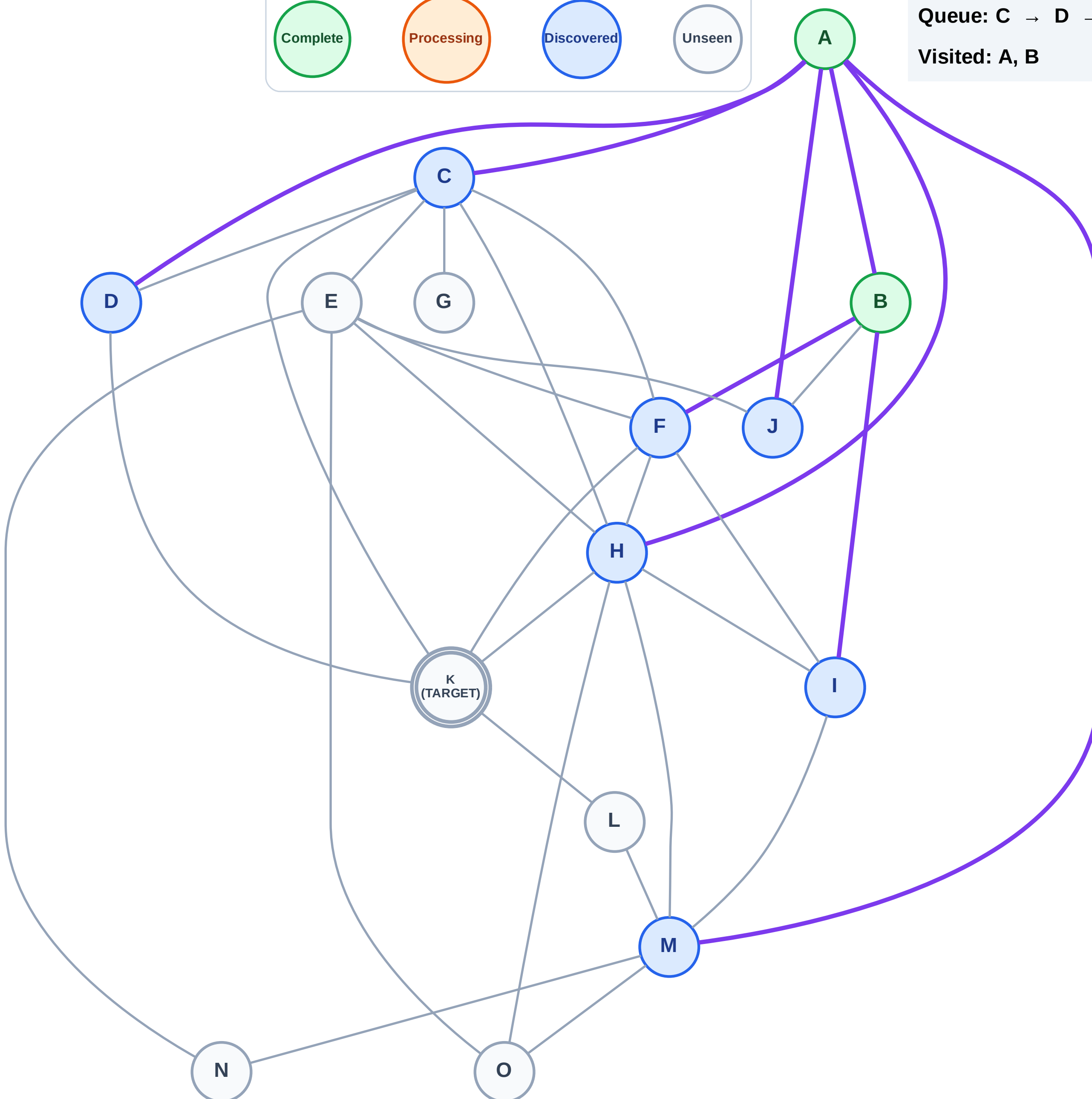
Discovered



Unseen

Queue: C → D → H → J → M → F → I

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

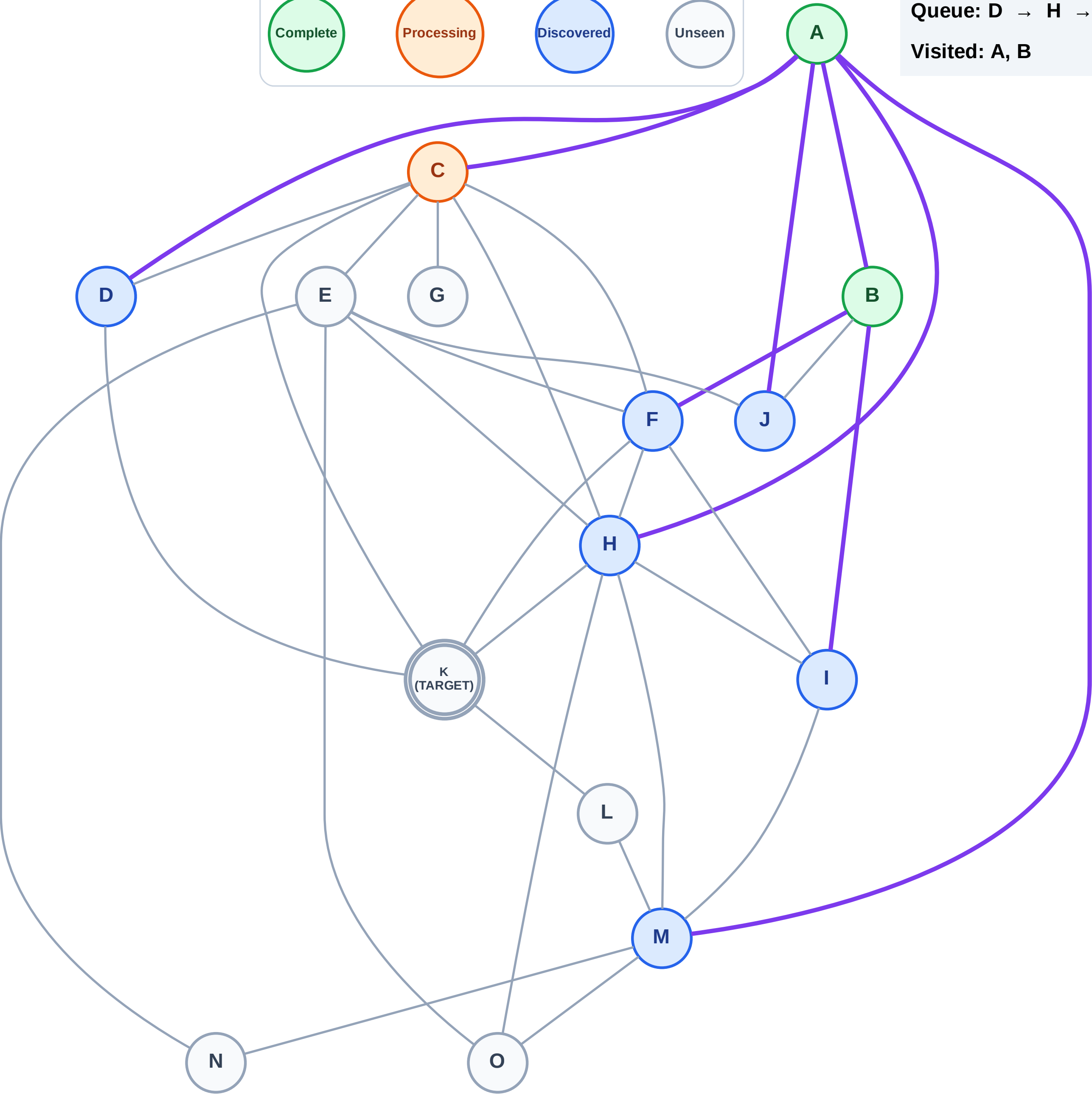
Processing

Discovered

Unseen

Queue: D → H → J → M → F → I

Visited: A, B



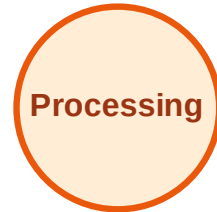
BFS Traversal

Step 7: Discover E, G, K from C

Legend



Complete



Processing



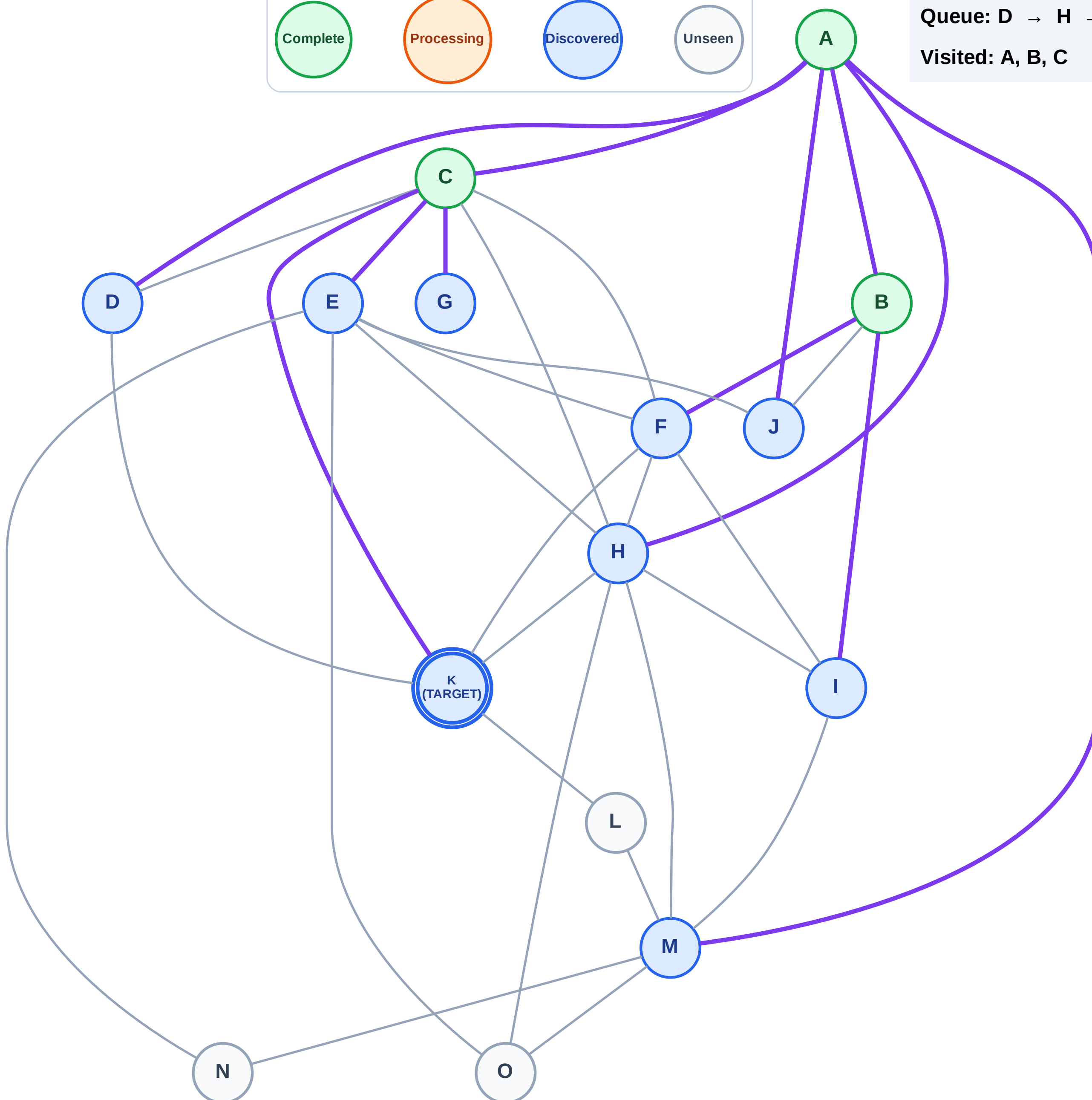
Discovered



Unseen

Queue: D → H → J → M → F → I → E → G → K

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

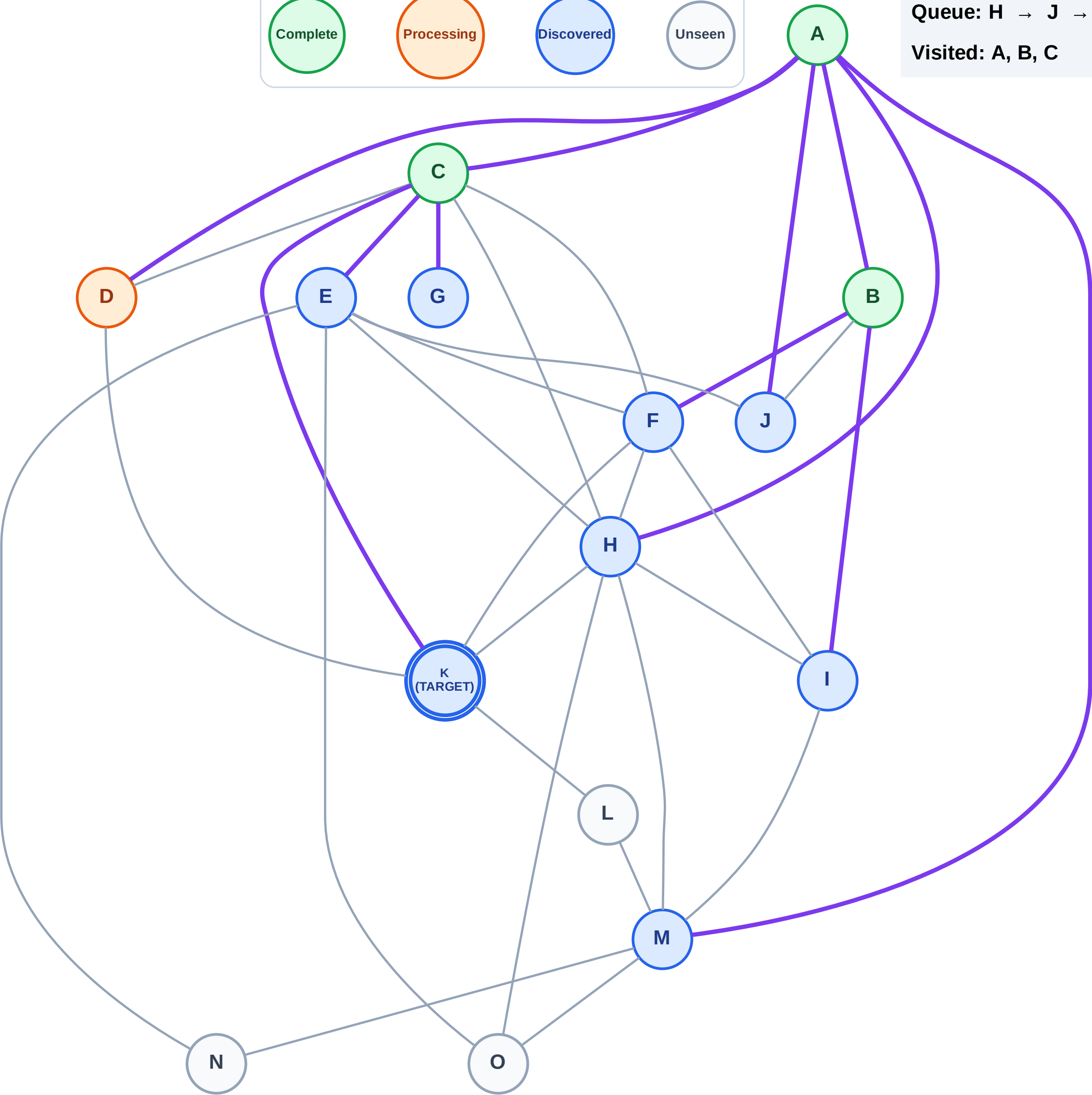
Processing

Discovered

Unseen

Queue: H → J → M → F → I → E → G → K

Visited: A, B, C



BFS Traversal

Step 9: No new neighbors from D

Legend



Complete



Processing



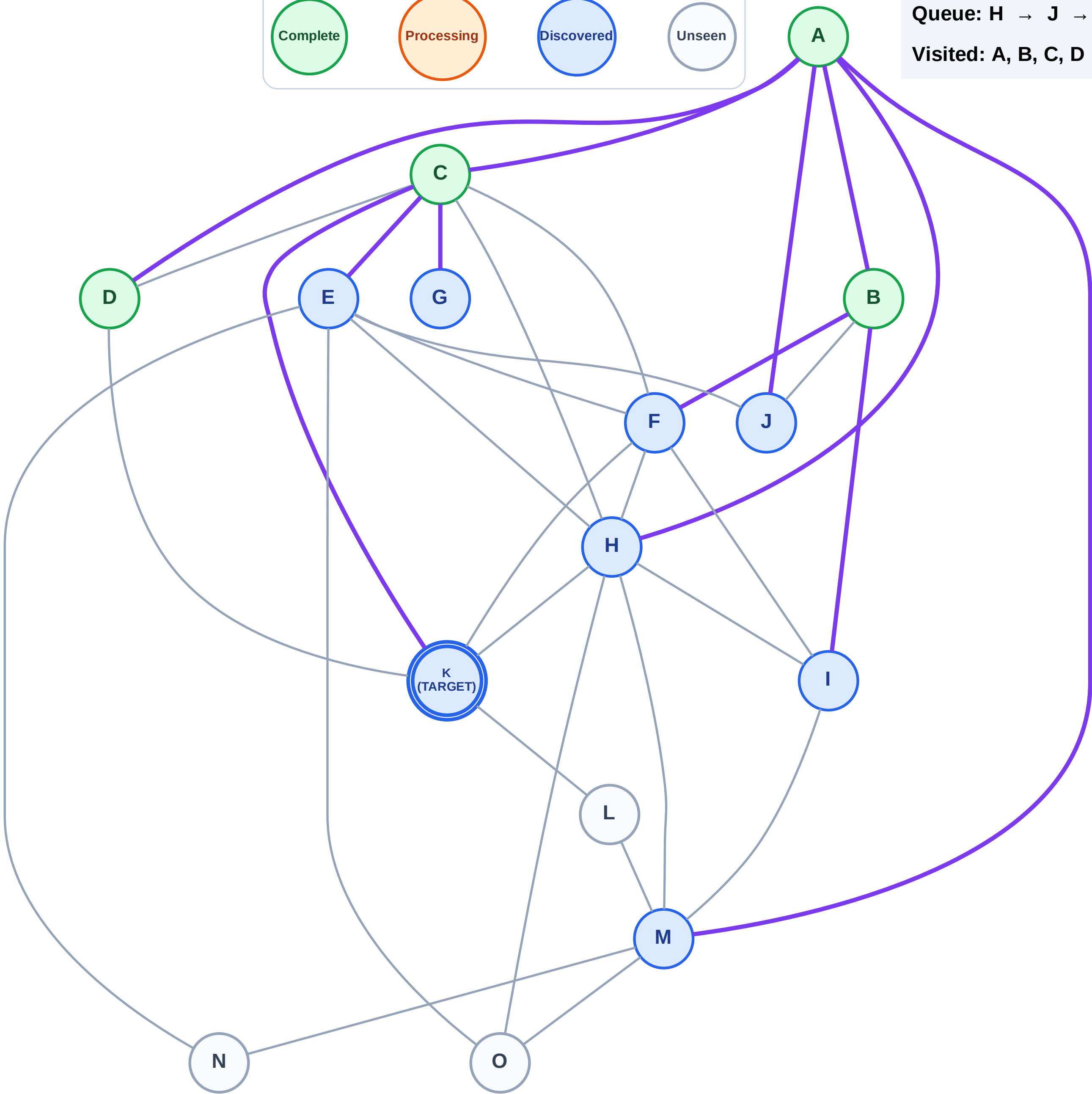
Discovered



Unseen

Queue: H → J → M → F → I → E → G → K

Visited: A, B, C, D



BFS Traversal

Step 10: Process node H

Legend

Complete

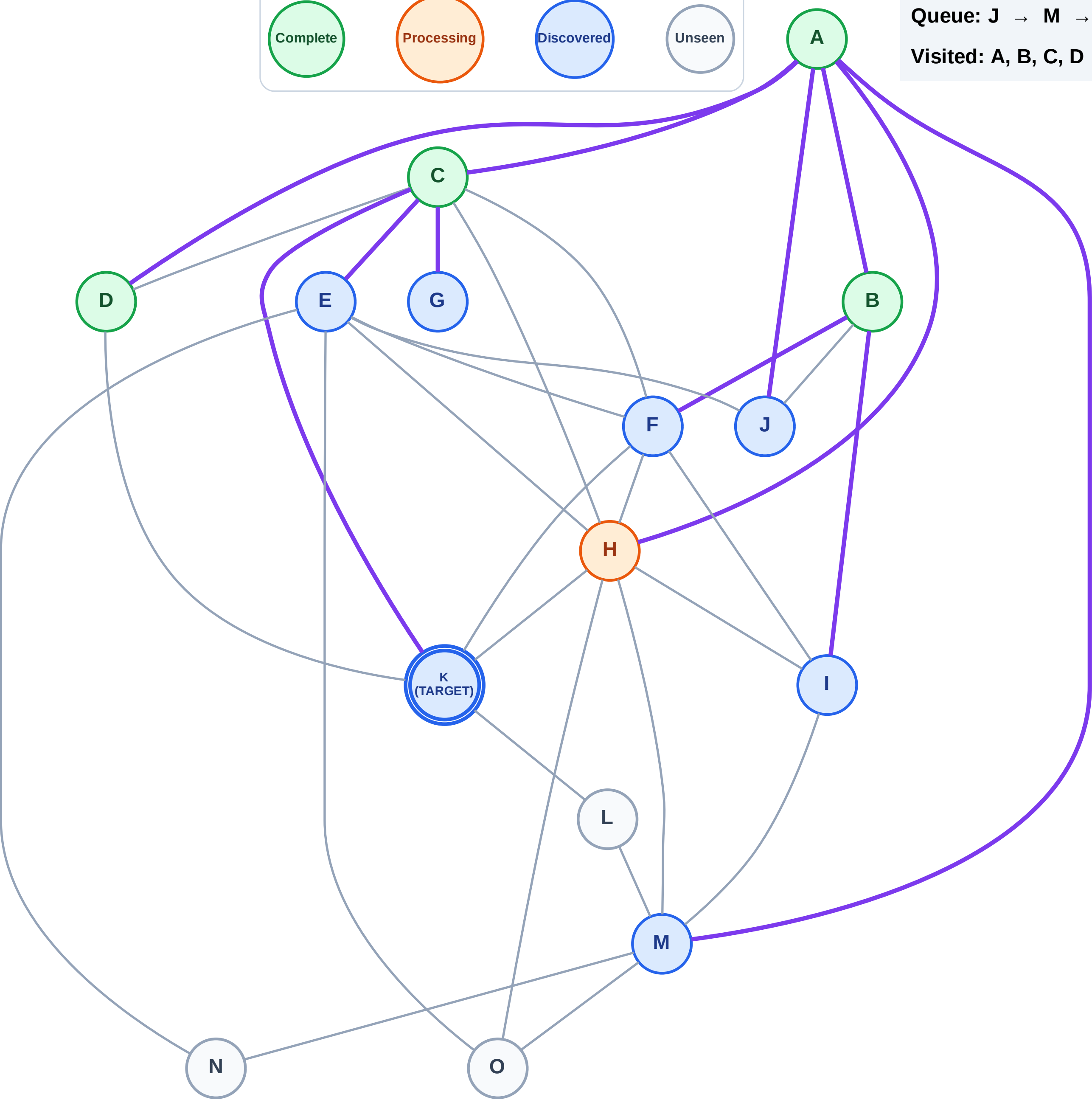
Processing

Discovered

Unseen

Queue: J → M → F → I → E → G → K

Visited: A, B, C, D



BFS Traversal

Step 11: Discover O from H

Legend



Complete



Processing



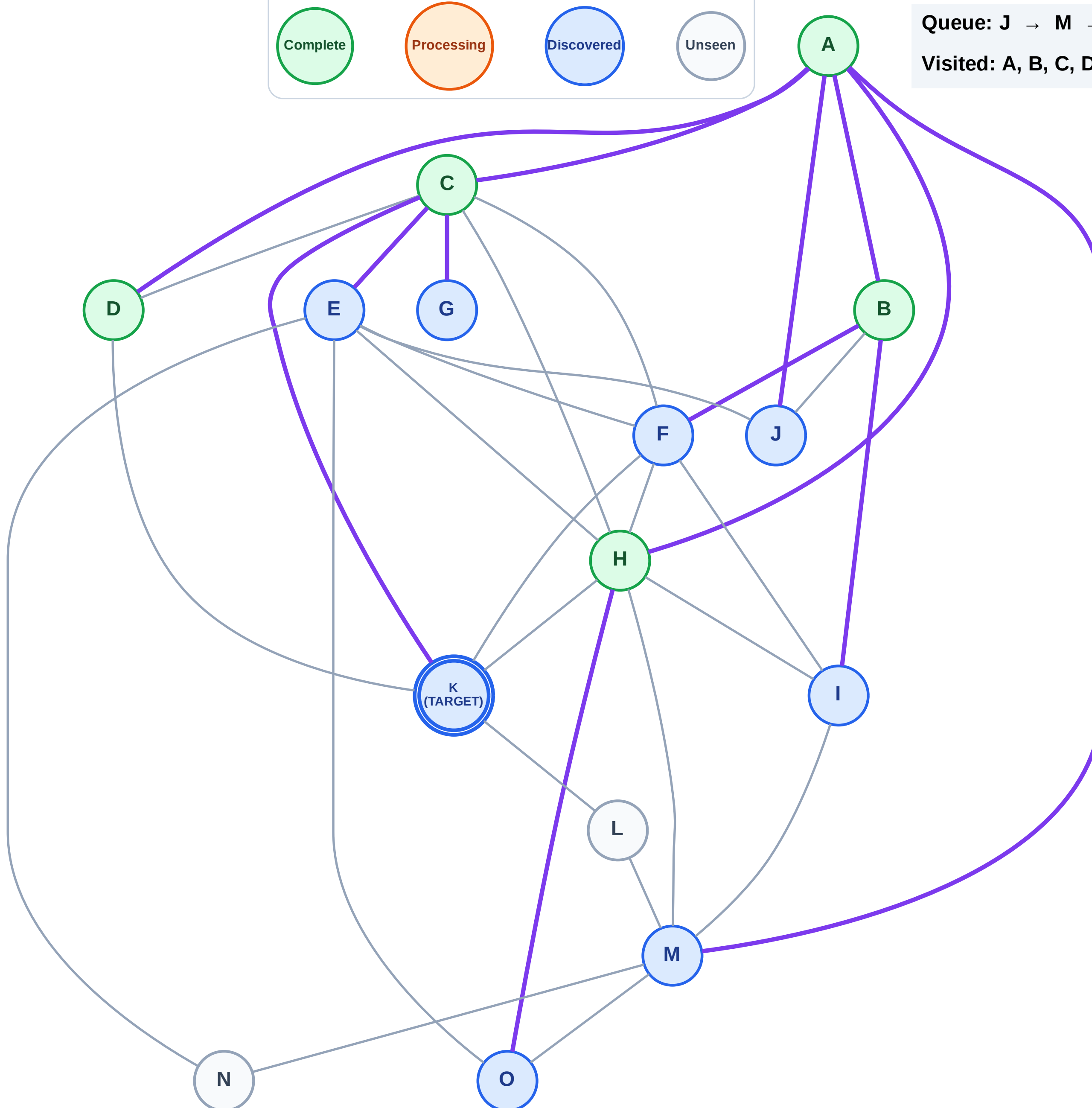
Discovered



Unseen

Queue: J → M → F → I → E → G → K → O

Visited: A, B, C, D, H



BFS Traversal

Step 12: Process node J

Legend

Complete

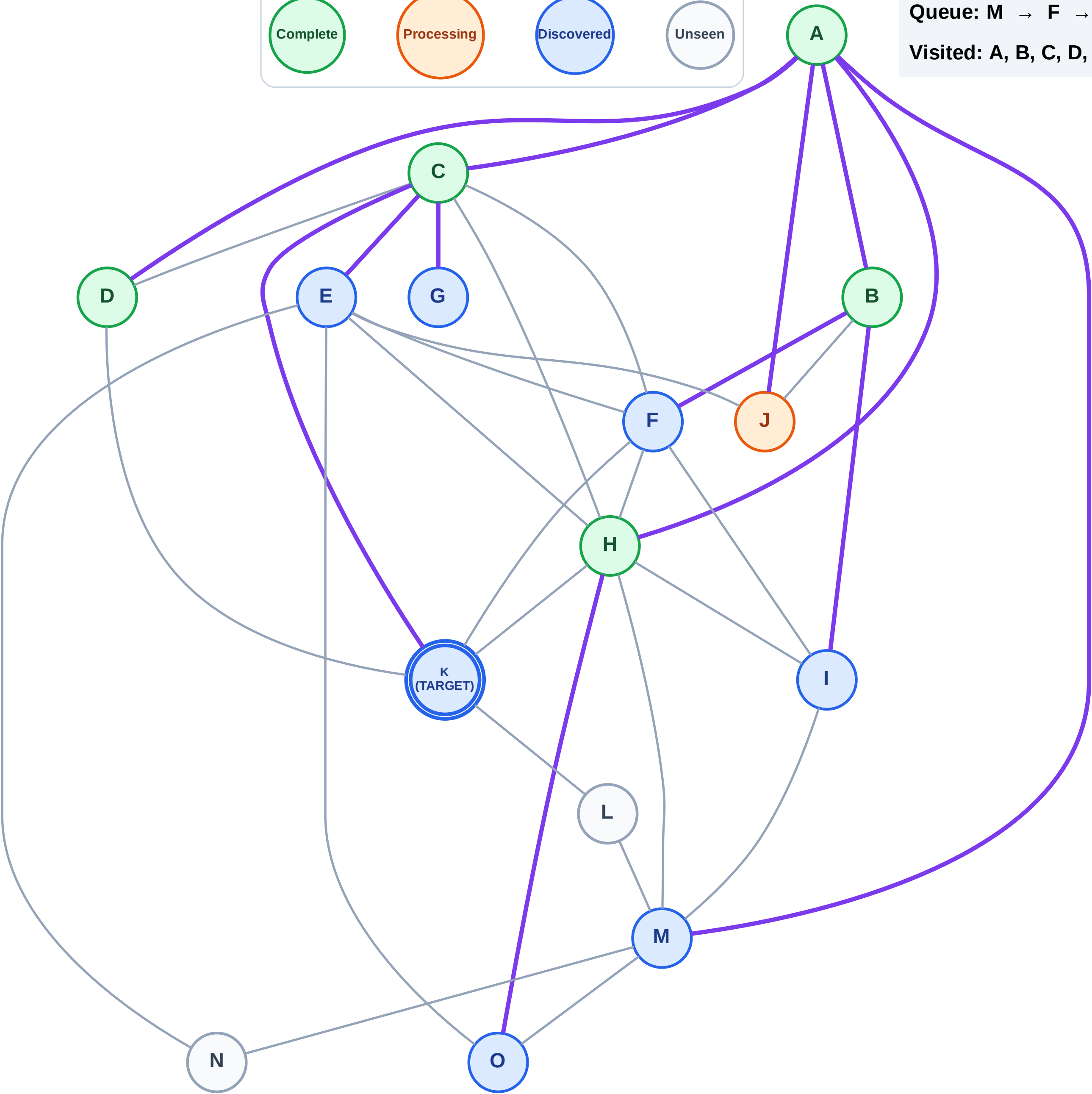
Processing

Discovered

Unseen

Queue: M → F → I → E → G → K → O

Visited: A, B, C, D, H



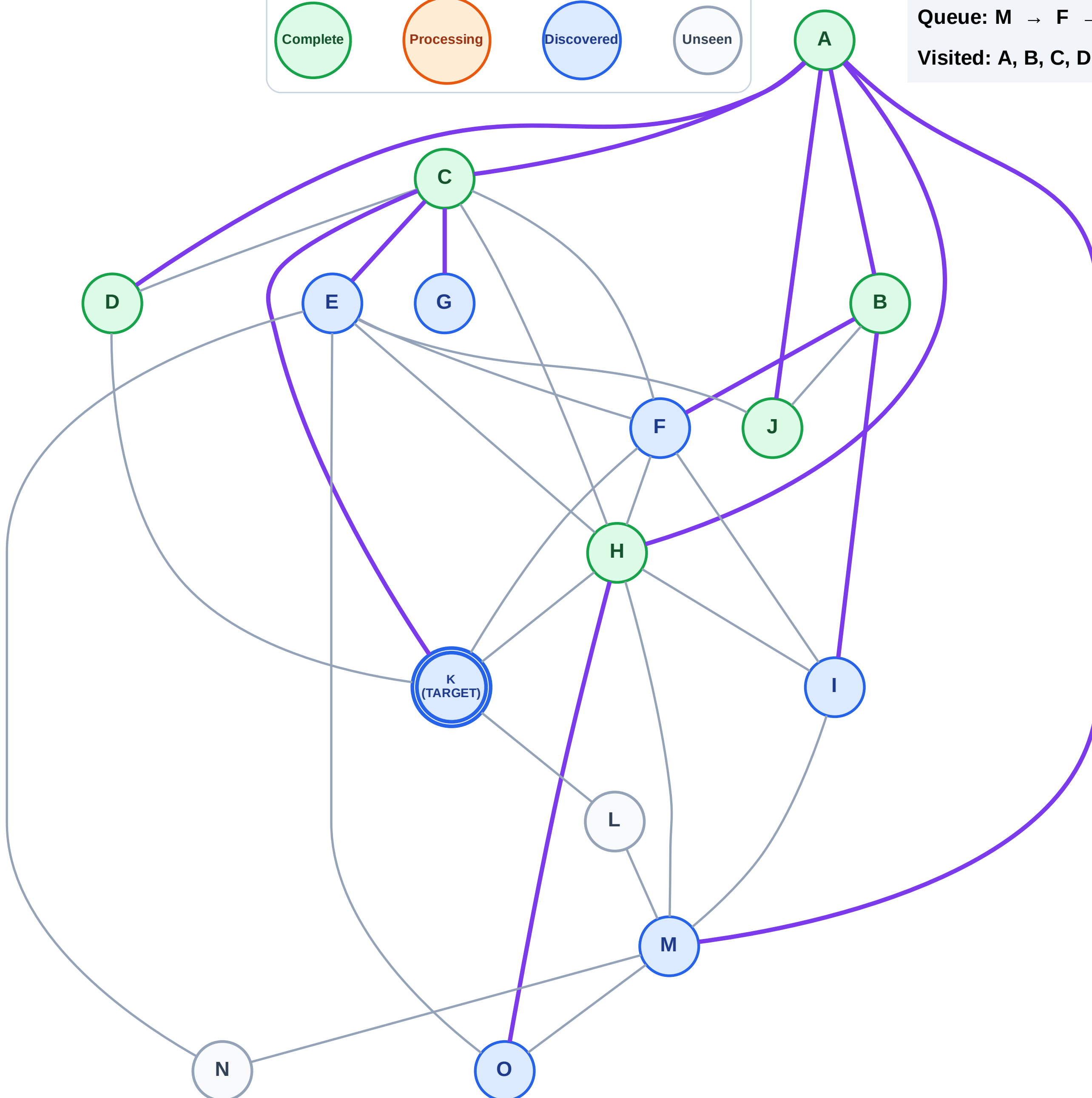
BFS Traversal

Step 13: No new neighbors from J

Legend



Queue: M → F → I → E → G → K → O
Visited: A, B, C, D, H, J



BFS Traversal

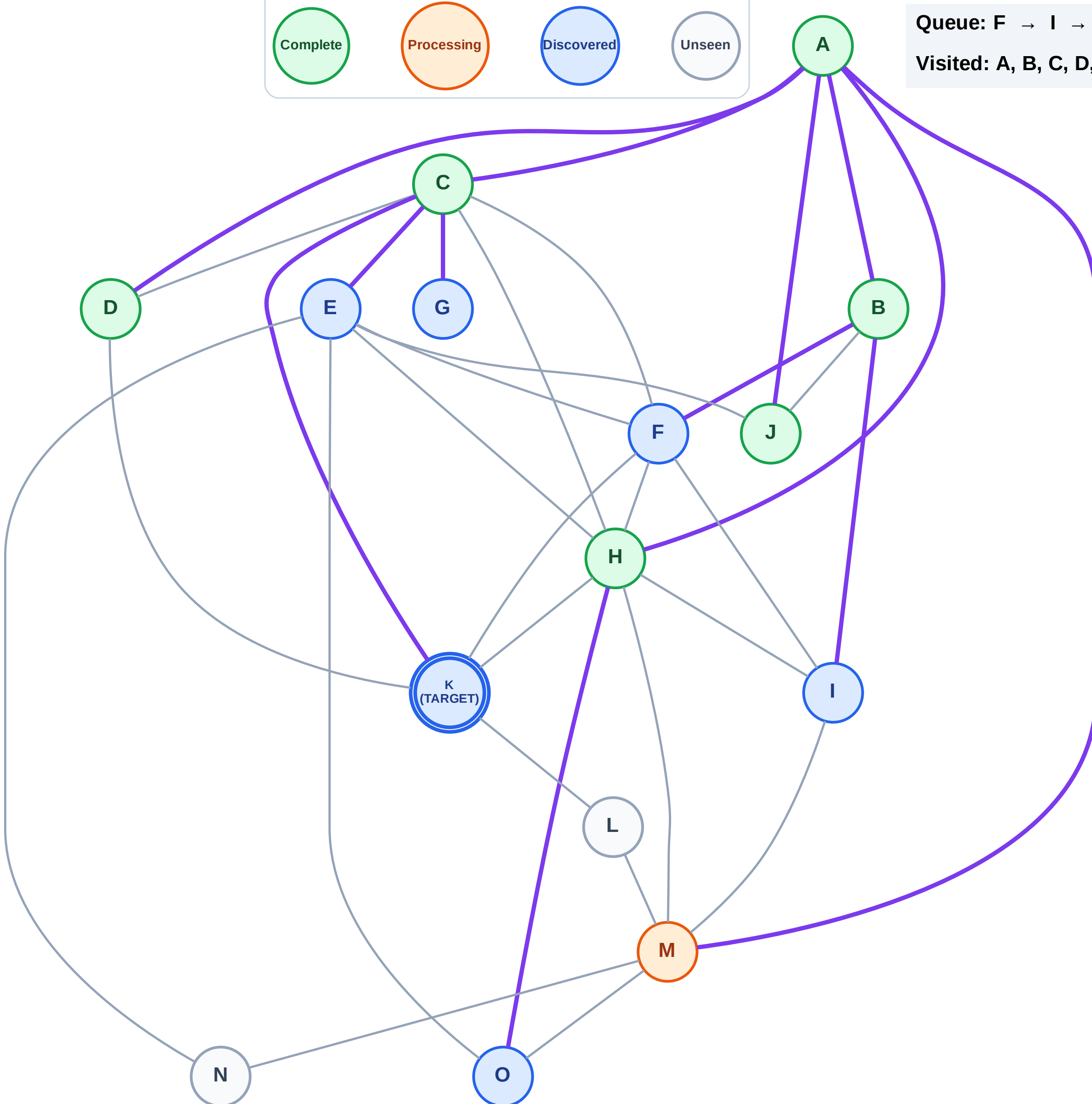
Step 14: Process node M

Legend



Queue: F → I → E → G → K → O

Visited: A, B, C, D, H, J



BFS Traversal

Step 15: Discover L, N from M

Legend

Complete

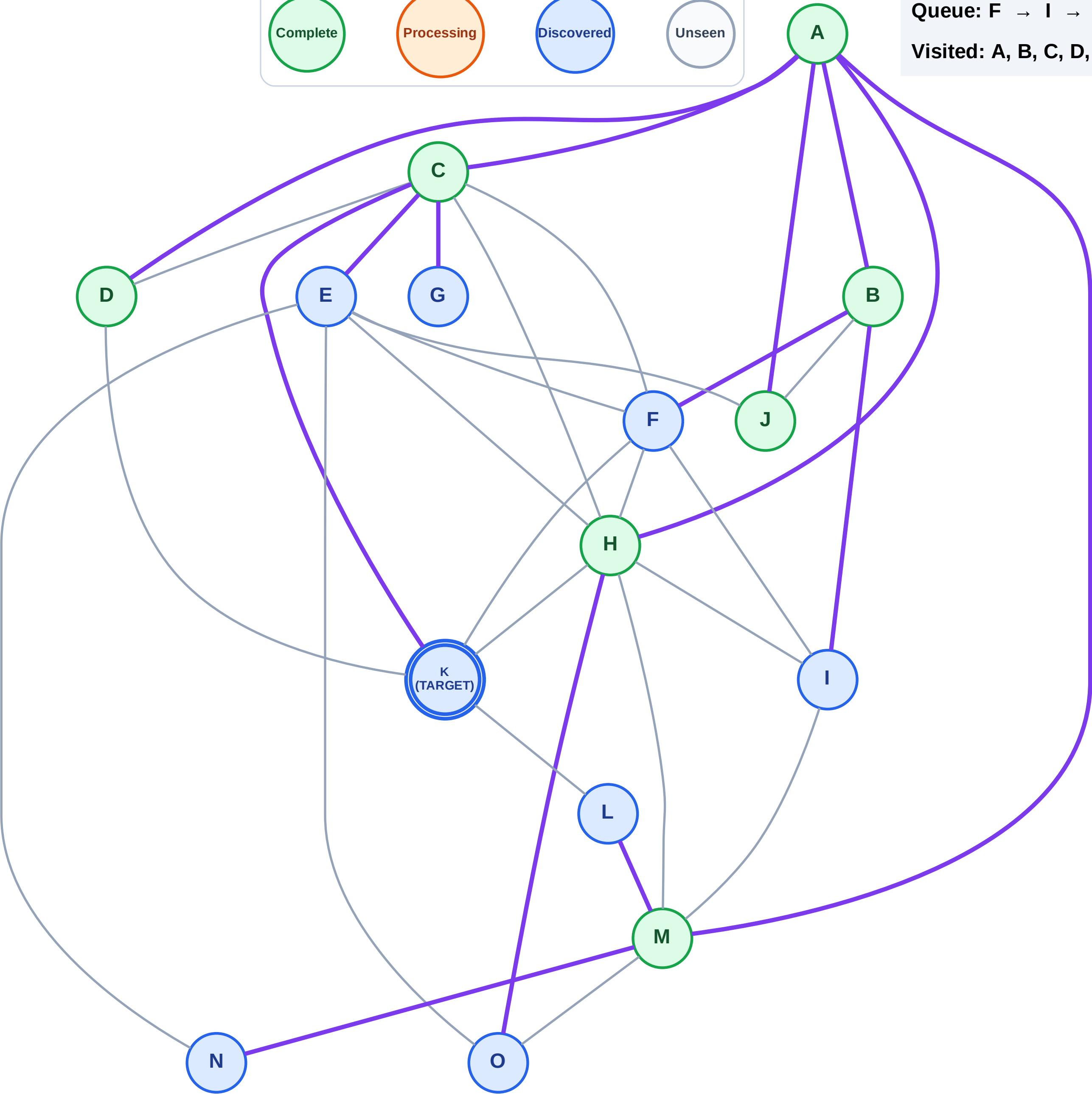
Processing

Discovered

Unseen

Queue: F → I → E → G → K → O → L → N

Visited: A, B, C, D, H, J, M



BFS Traversal

Step 16: Process node F

Legend

Complete

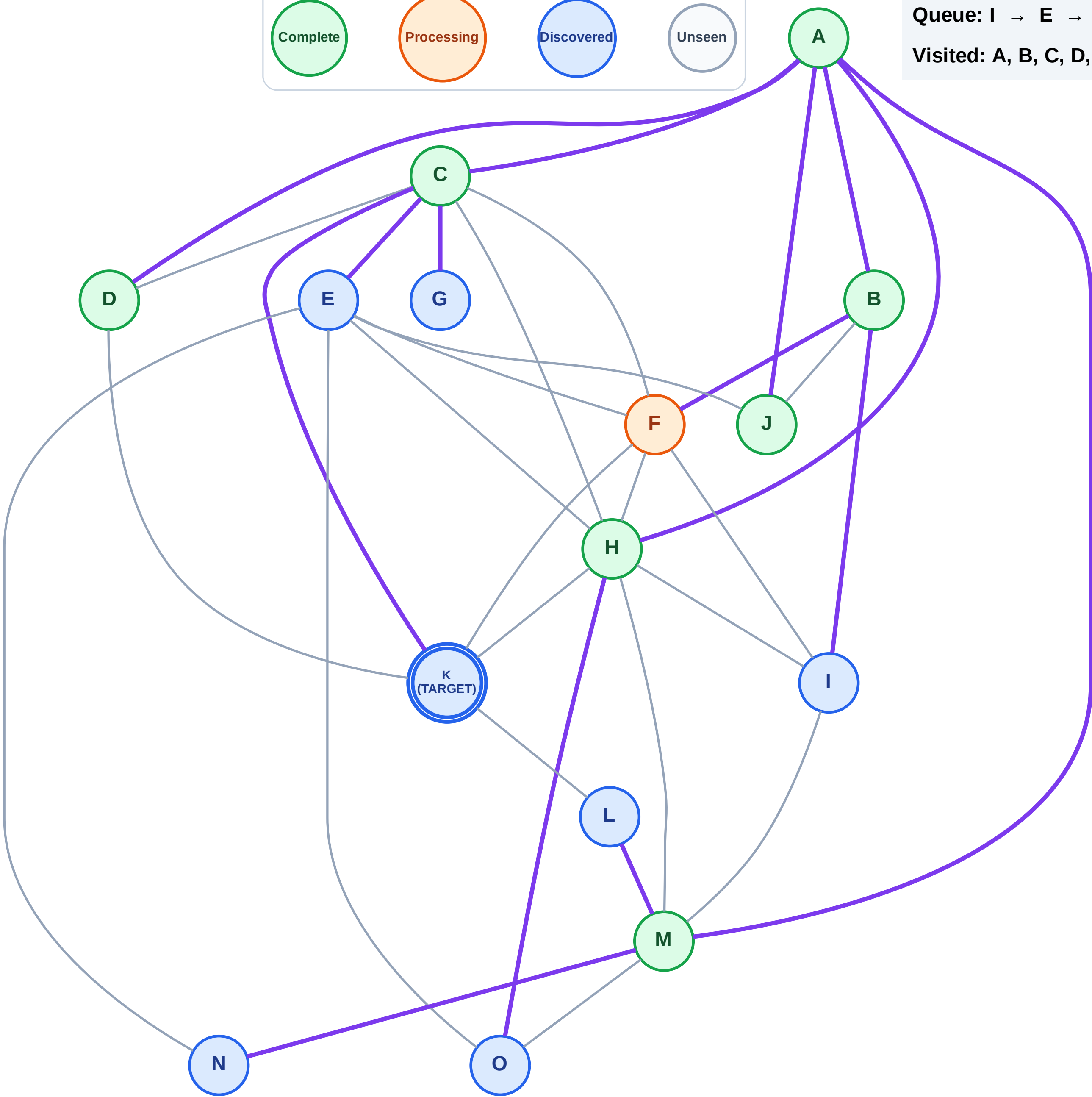
Processing

Discovered

Unseen

Queue: I → E → G → K → O → L → N

Visited: A, B, C, D, H, J, M



BFS Traversal

Step 17: No new neighbors from F

Legend

Complete

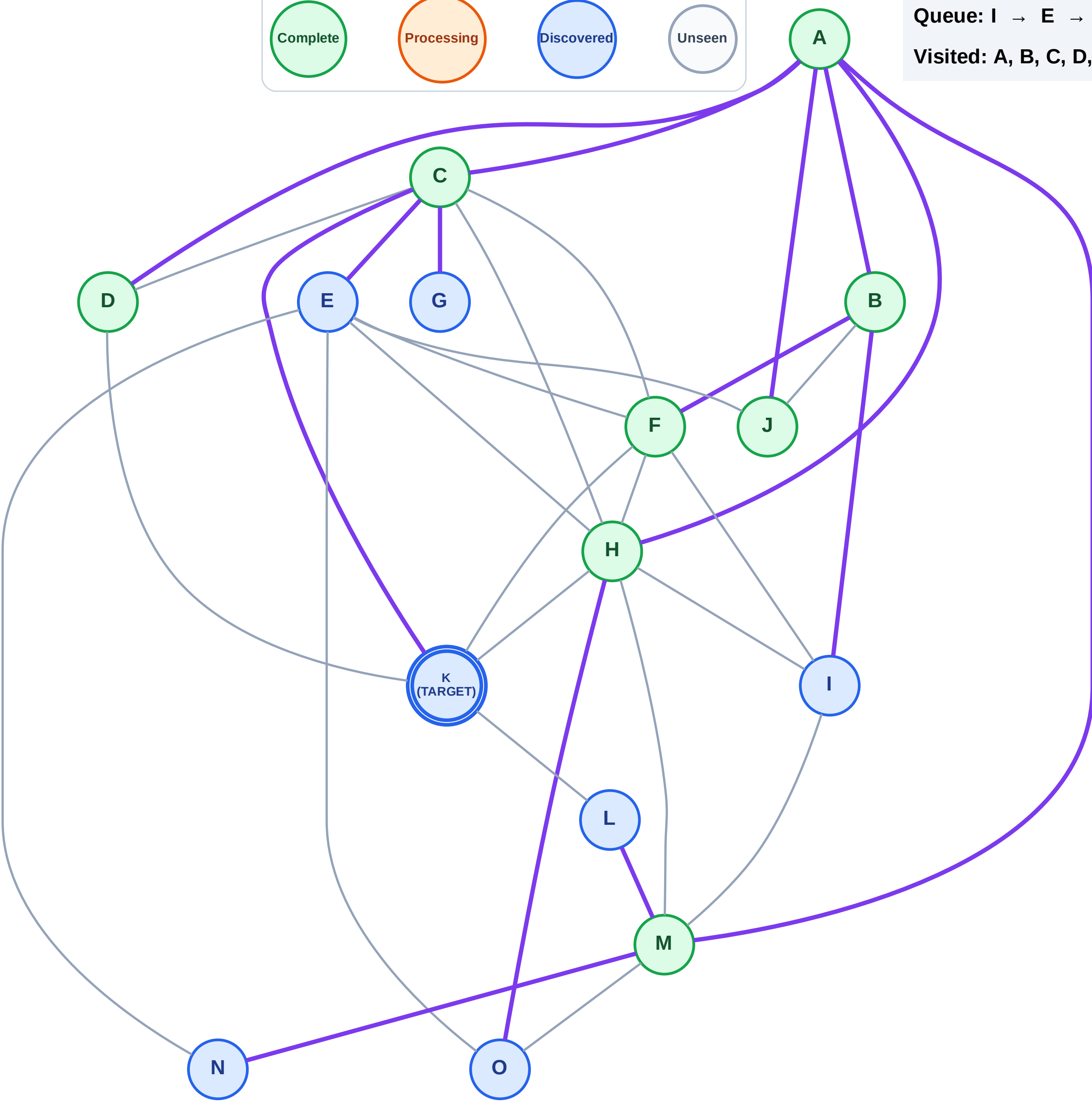
Processing

Discovered

Unseen

Queue: I → E → G → K → O → L → N

Visited: A, B, C, D, F, H, J, M



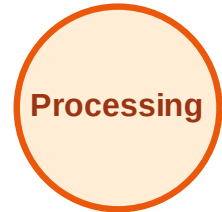
BFS Traversal

Step 18: Process node I

Legend



Complete



Processing



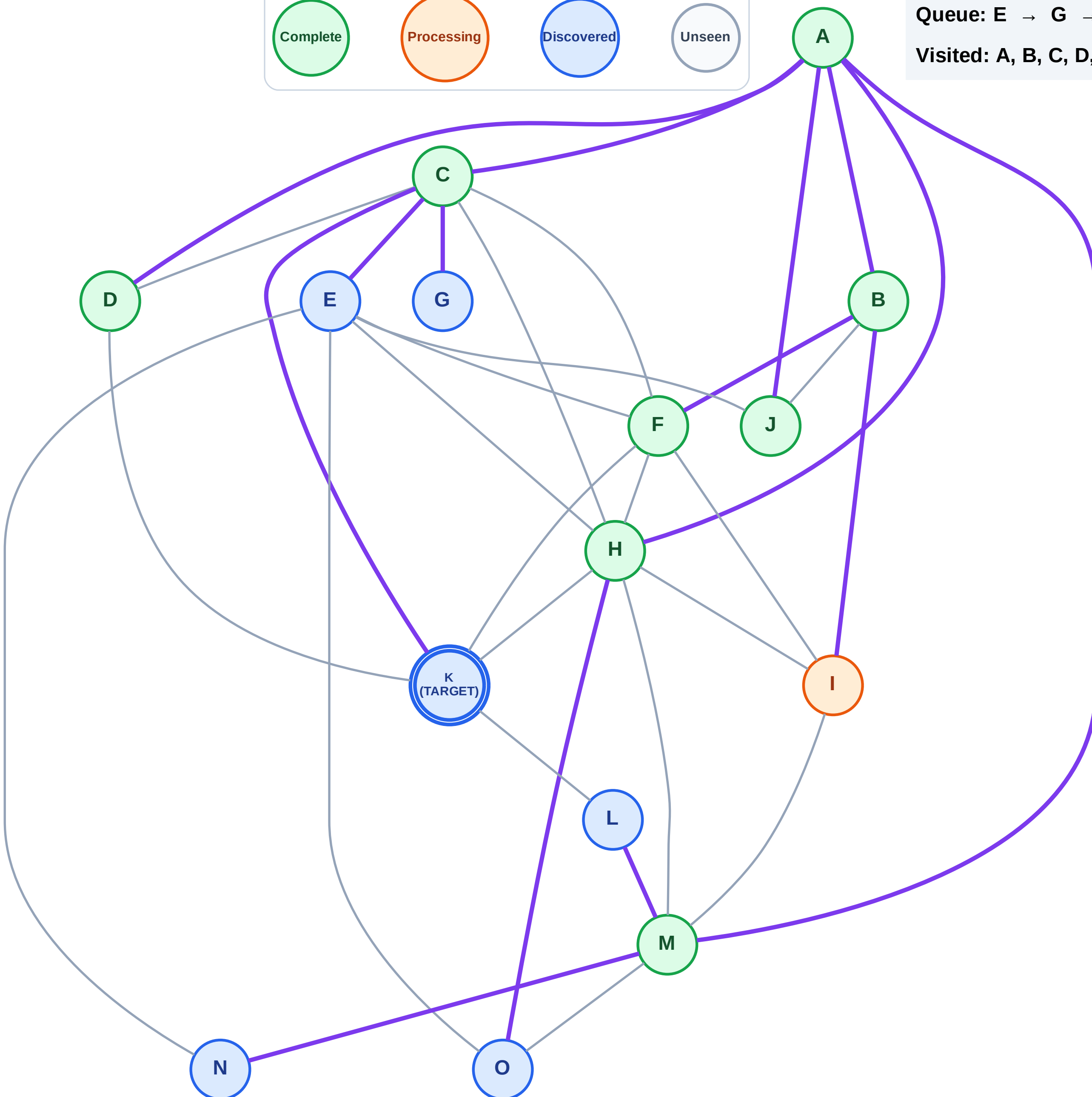
Discovered



Unseen

Queue: E → G → K → O → L → N

Visited: A, B, C, D, F, H, J, M



BFS Traversal

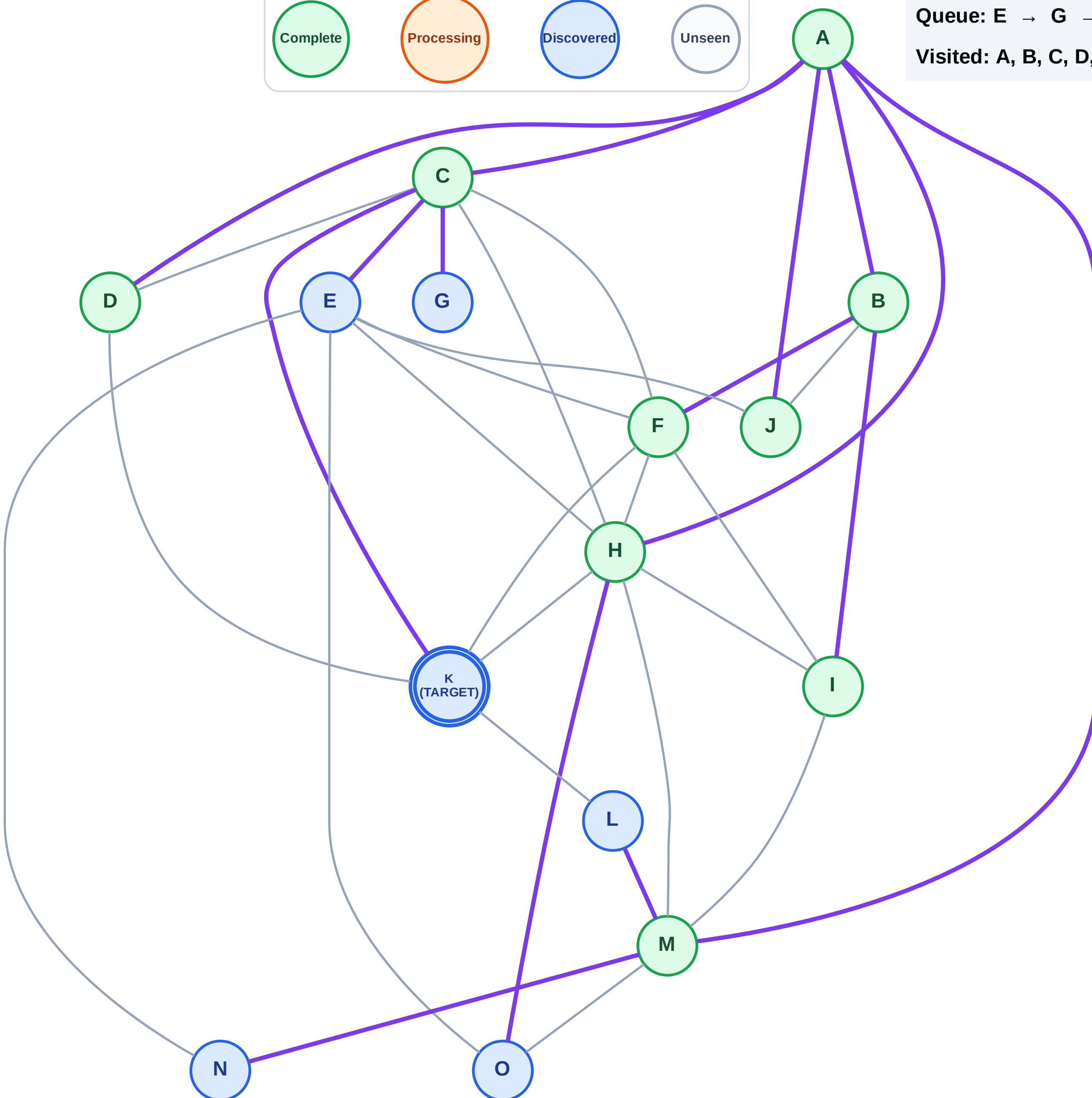
Step 19: No new neighbors from I

Legend



Queue: E → G → K → O → L → N

Visited: A, B, C, D, F, H, I, J, M



BFS Traversal

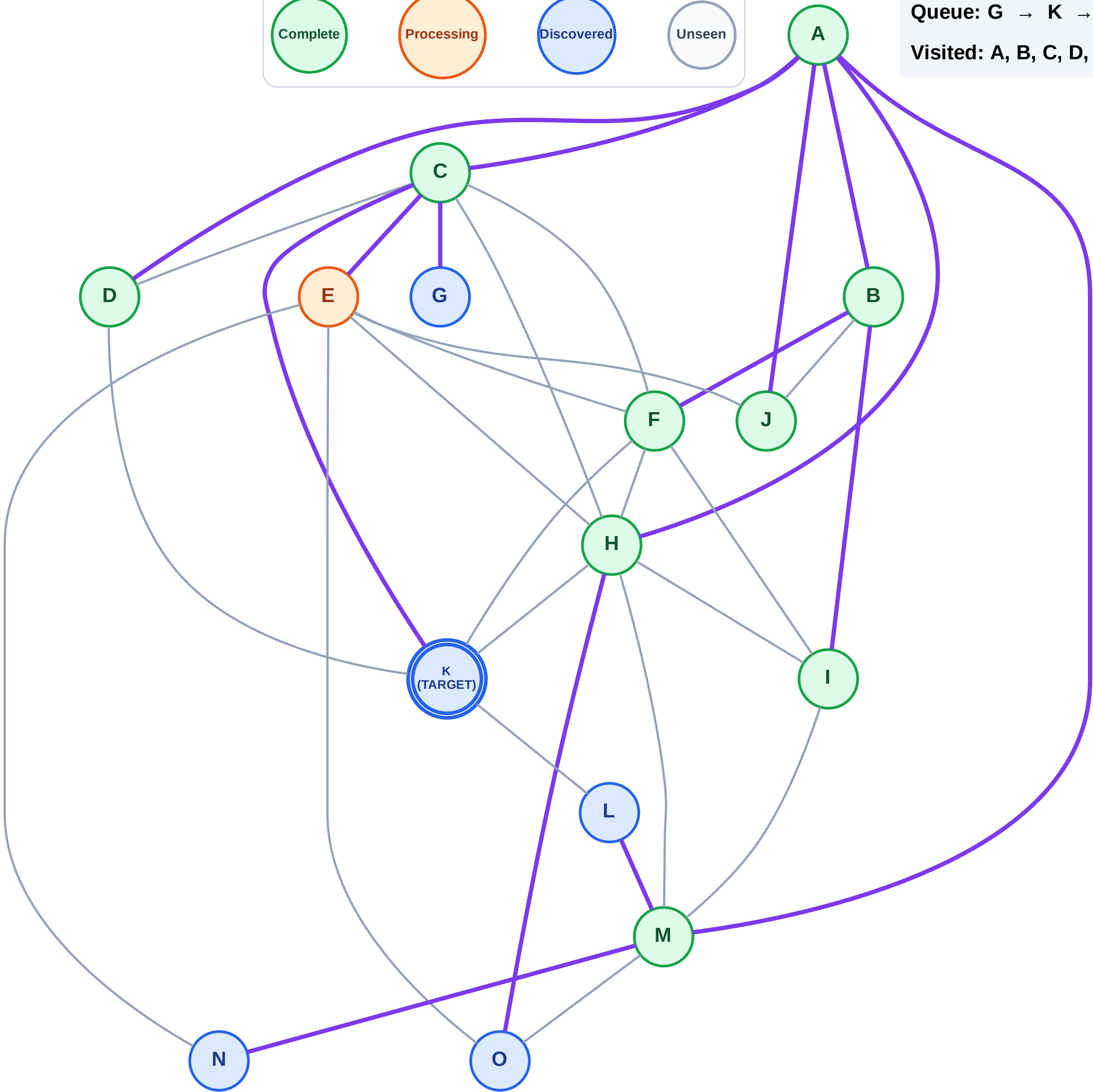
Step 20: Process node E

Legend



Queue: G → K → O → L → N

Visited: A, B, C, D, F, H, I, J, M



BFS Traversal

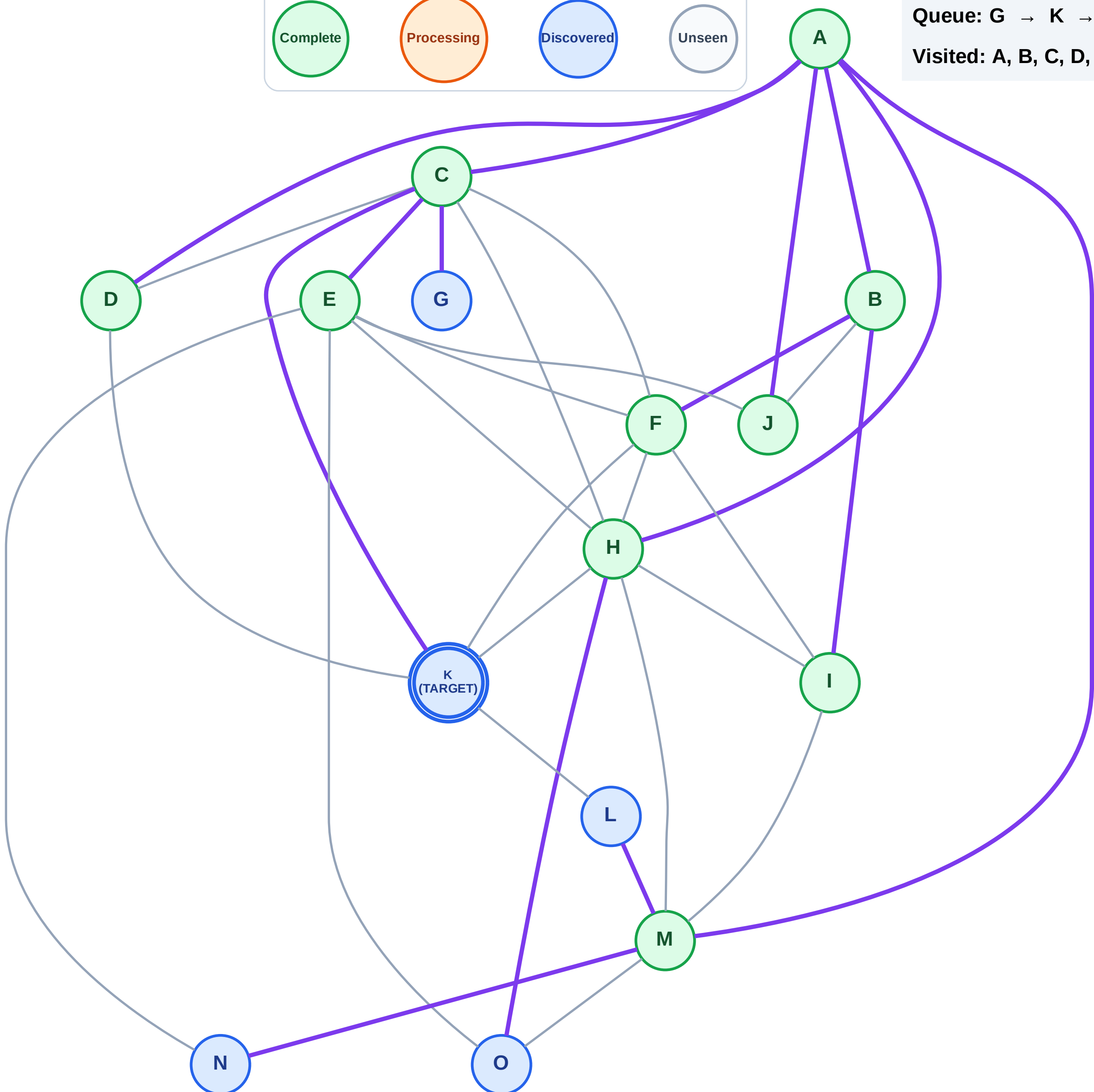
Step 21: No new neighbors from E

Legend



Queue: G → K → O → L → N

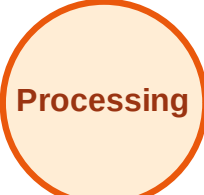
Visited: A, B, C, D, E, F, H, I, J, M



BFS Traversal

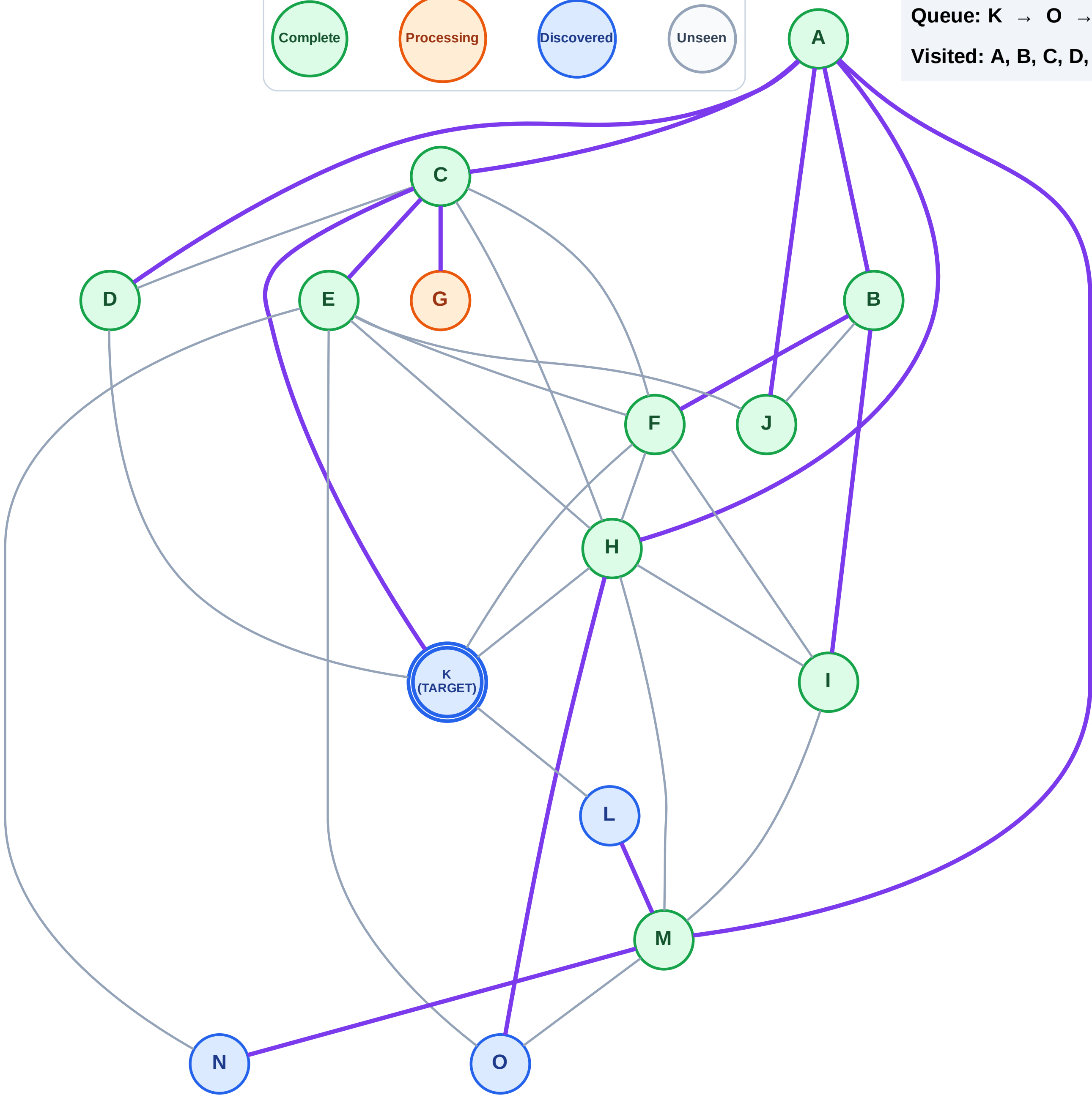
Step 22: Process node G

Legend



Queue: K → O → L → N

Visited: A, B, C, D, E, F, H, I, J, M



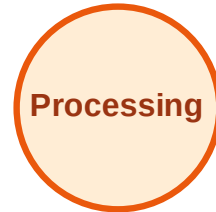
BFS Traversal

Step 23: No new neighbors from G

Legend



Complete



Processing

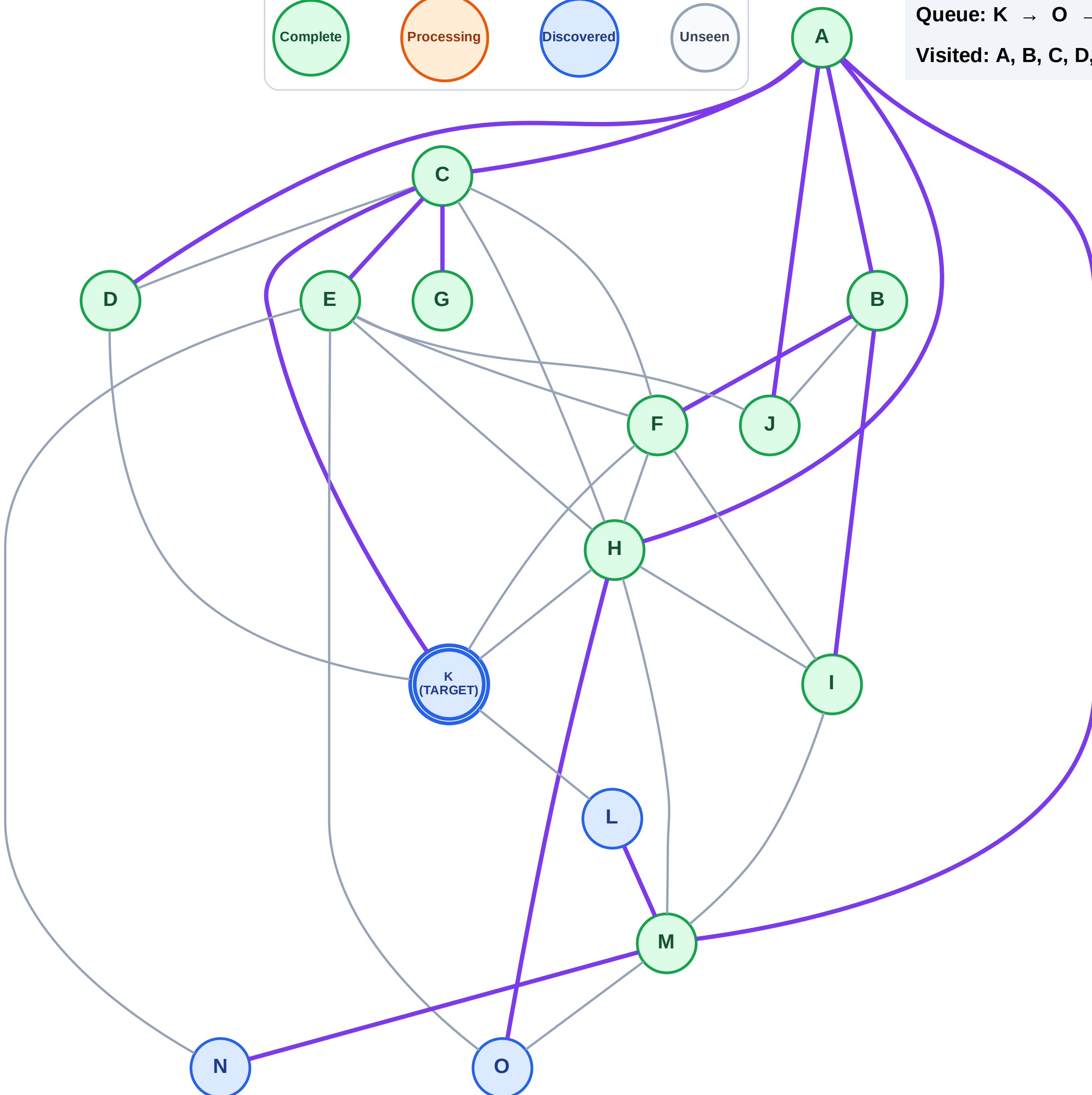


Discovered



Unseen

Queue: K → O → L → N
Visited: A, B, C, D, E, F, G, H, I, J, M



BFS Traversal

Step 24: Process node K

Legend

Complete

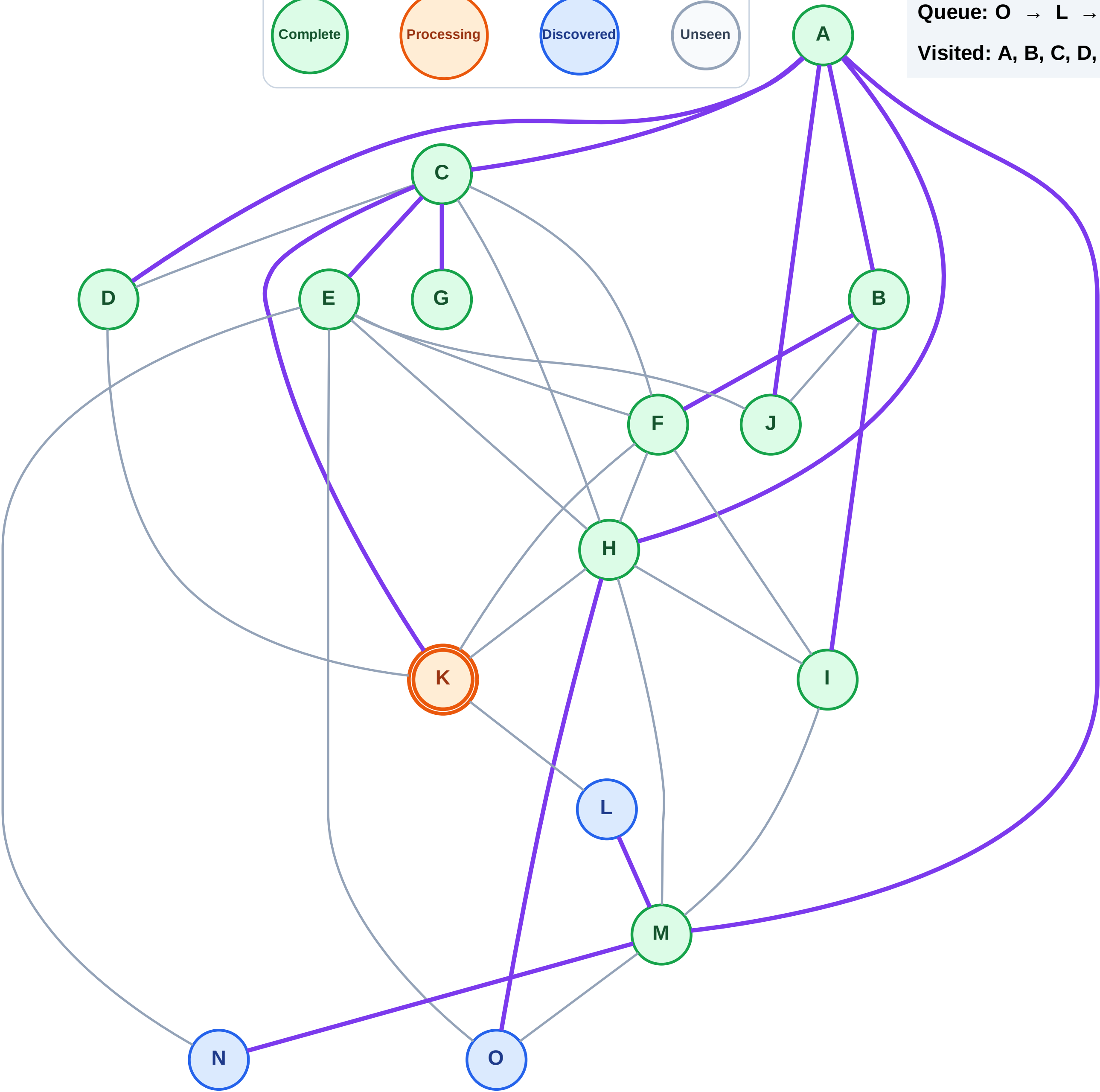
Processing

Discovered

Unseen

Queue: O → L → N

Visited: A, B, C, D, E, F, G, H, I, J, M



BFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend

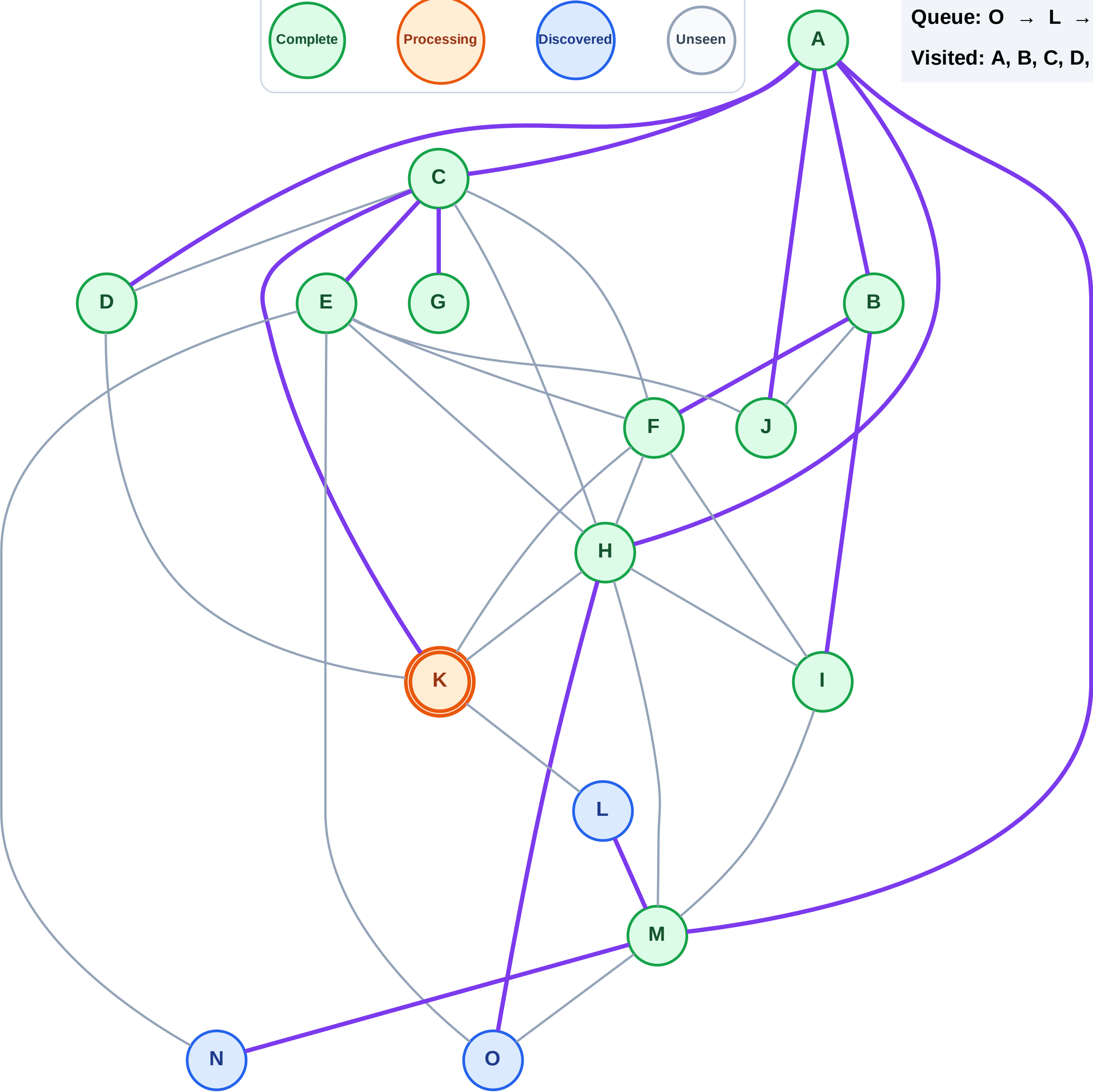
Complete

Processing

Discovered

Unseen

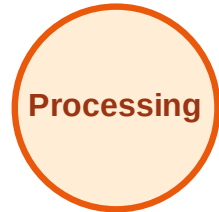
Queue: O → L → N
Visited: A, B, C, D, E, F, G, H, I, J, K, M



DFS Traversal

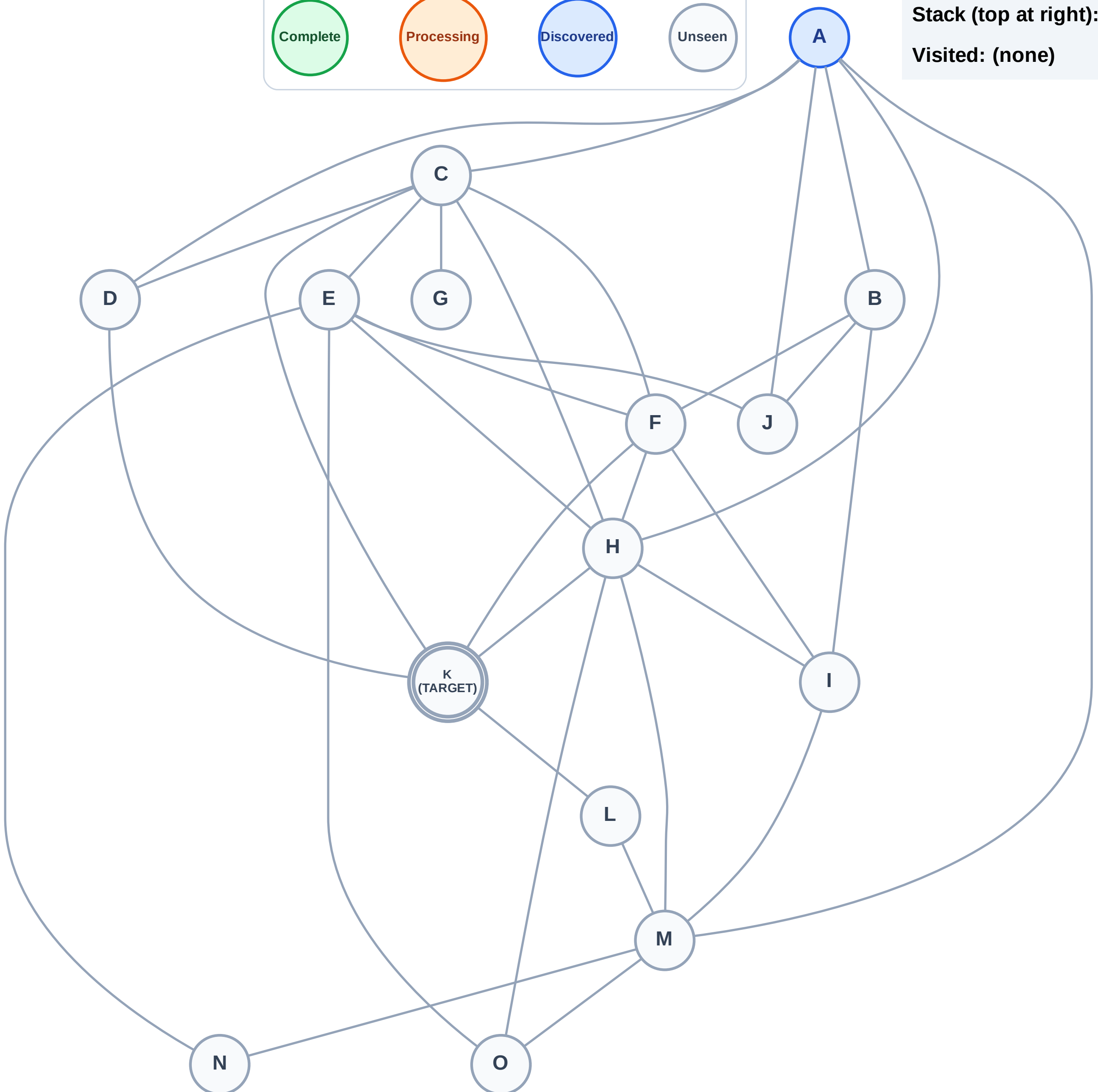
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

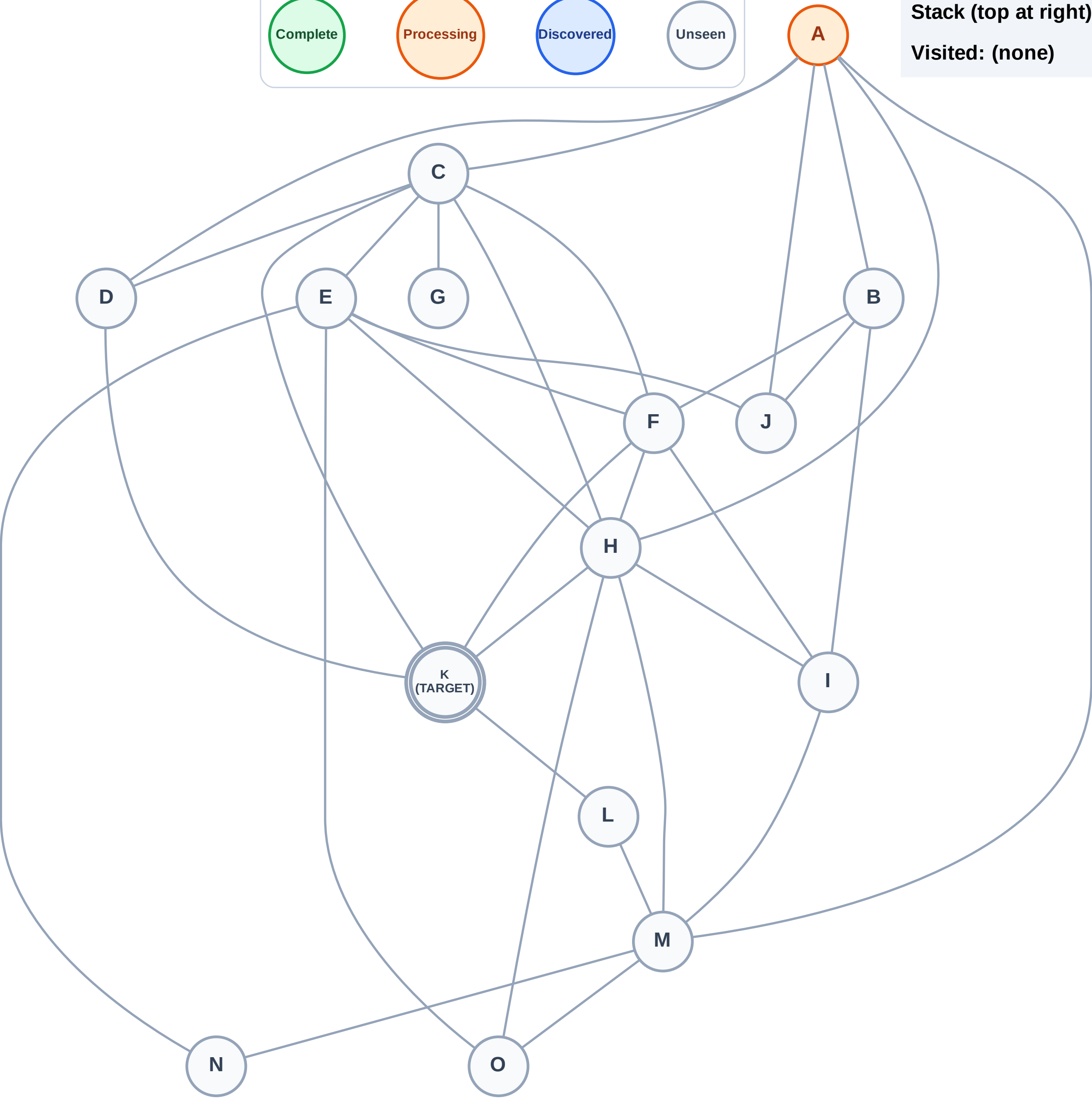
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover M, J, H, D, C, B from A

Legend

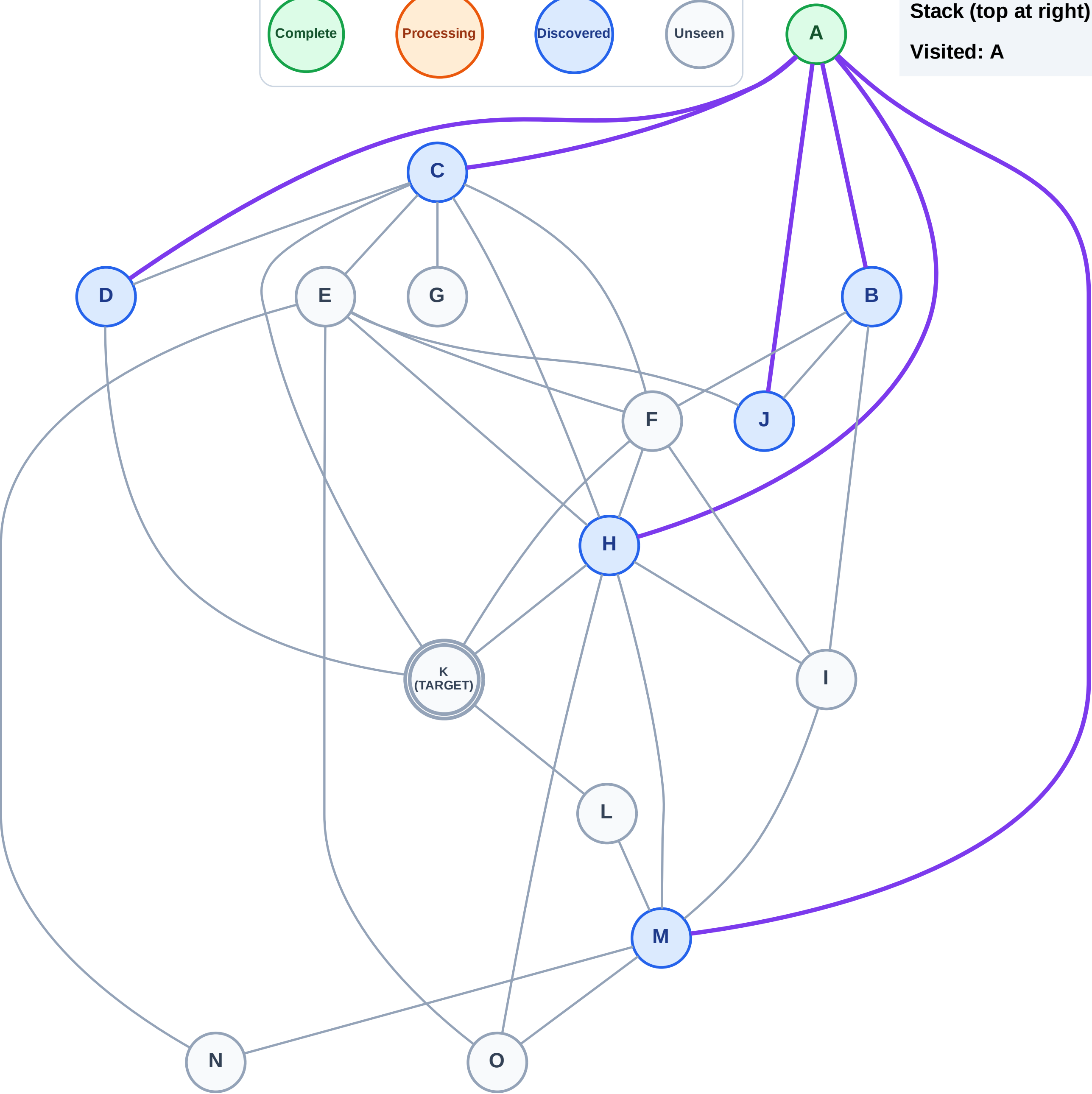
Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | H | D | C | B
Visited: A



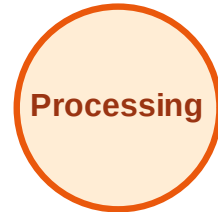
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



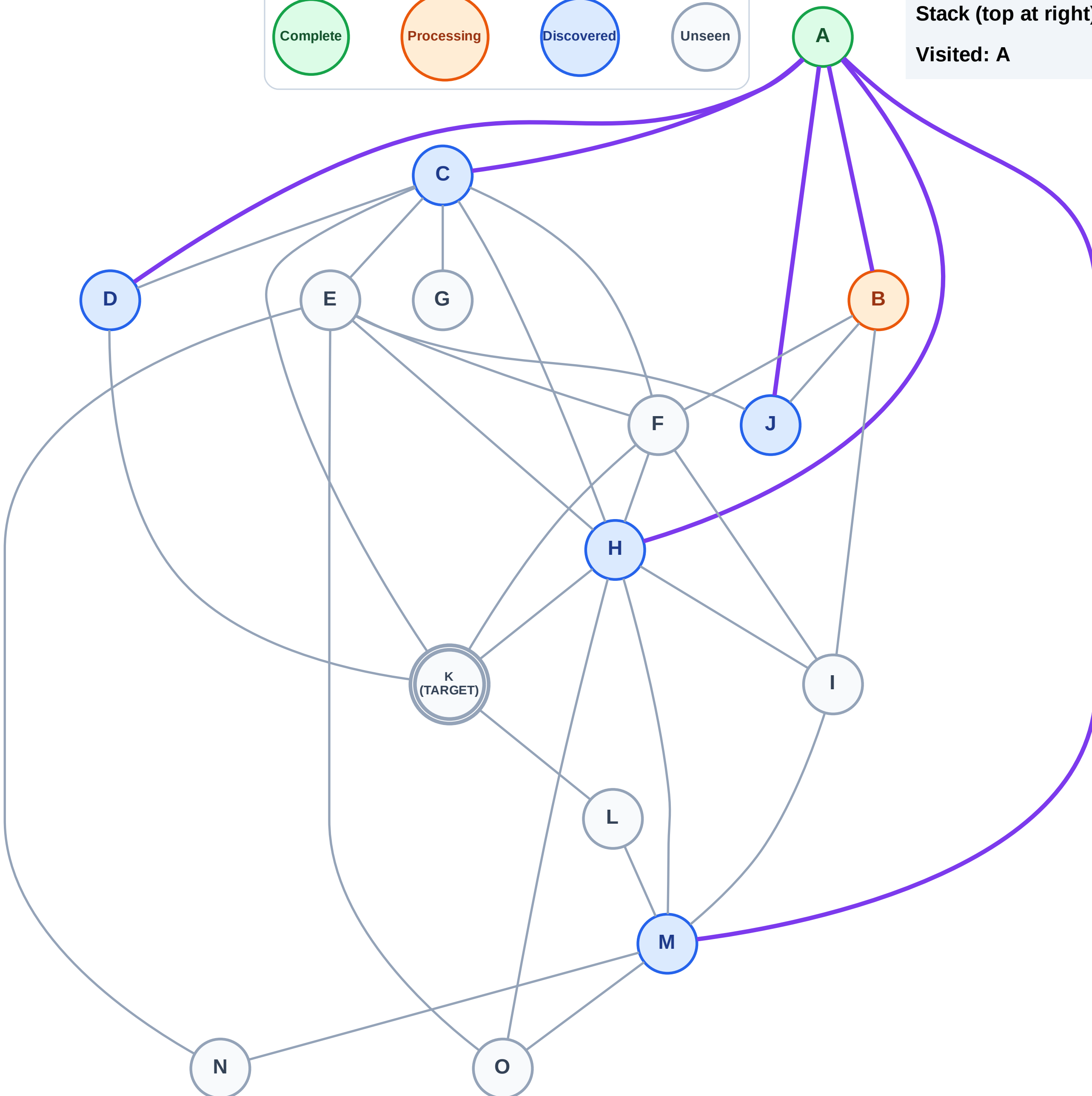
Discovered



Unseen

Stack (top at right): M | J | H | D | C

Visited: A



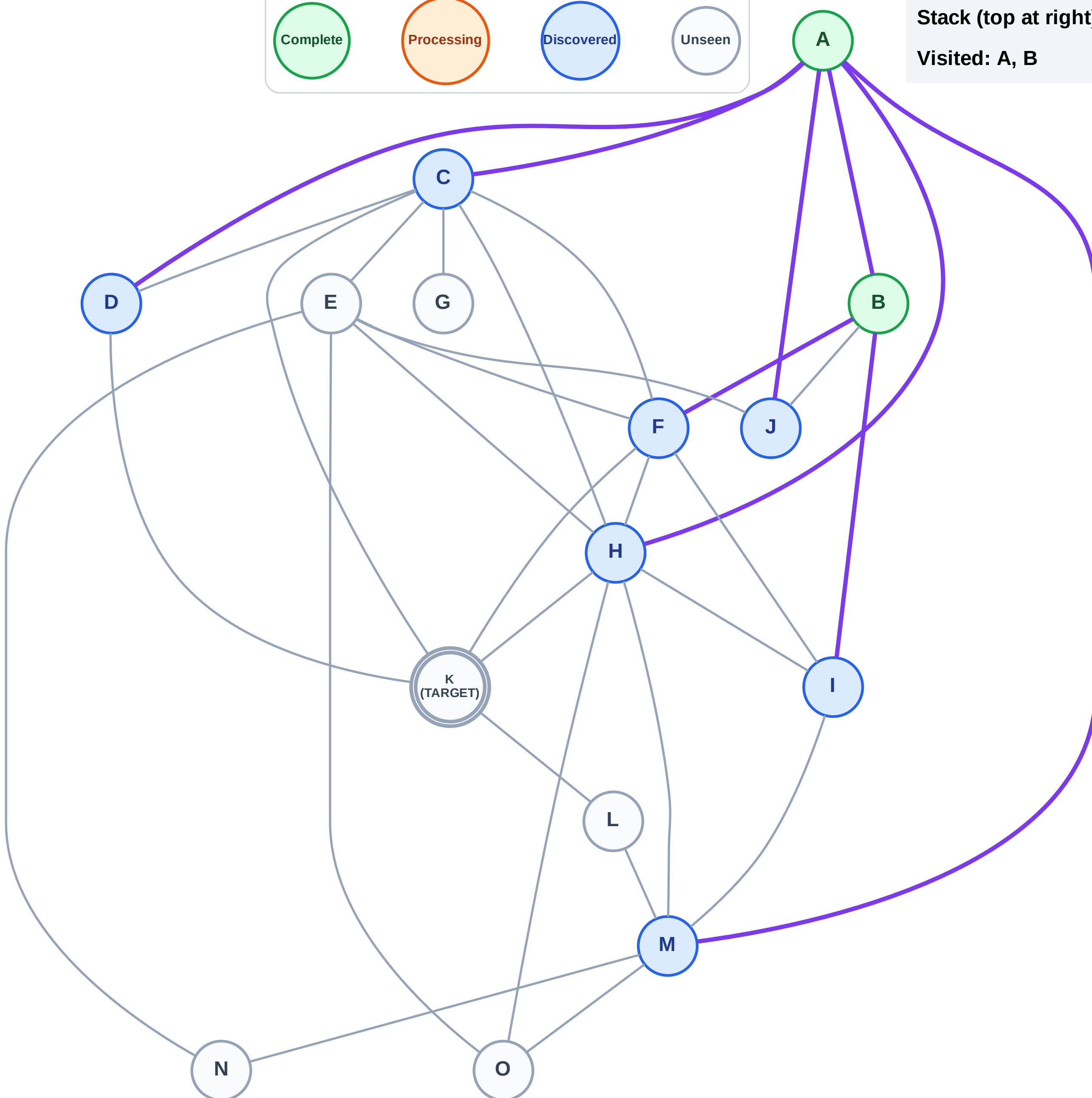
DFS Traversal

Step 5: Discover I, F from B

Legend



Stack (top at right): M | J | H | D | C | I | F
Visited: A, B



DFS Traversal

Step 6: Process node F

Legend

Complete

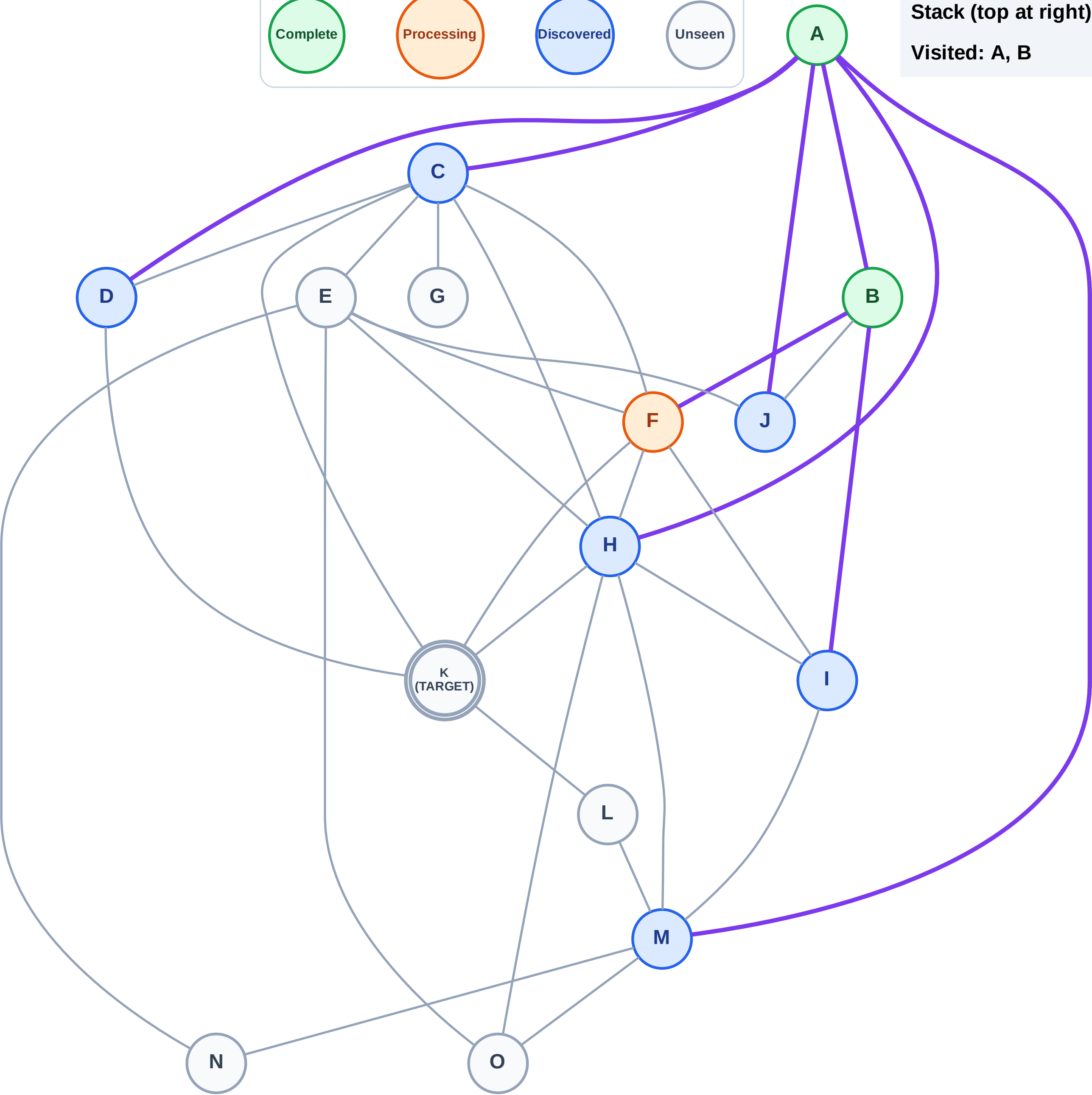
Processing

Discovered

Unseen

Stack (top at right): M | J | H | D | C | I

Visited: A, B

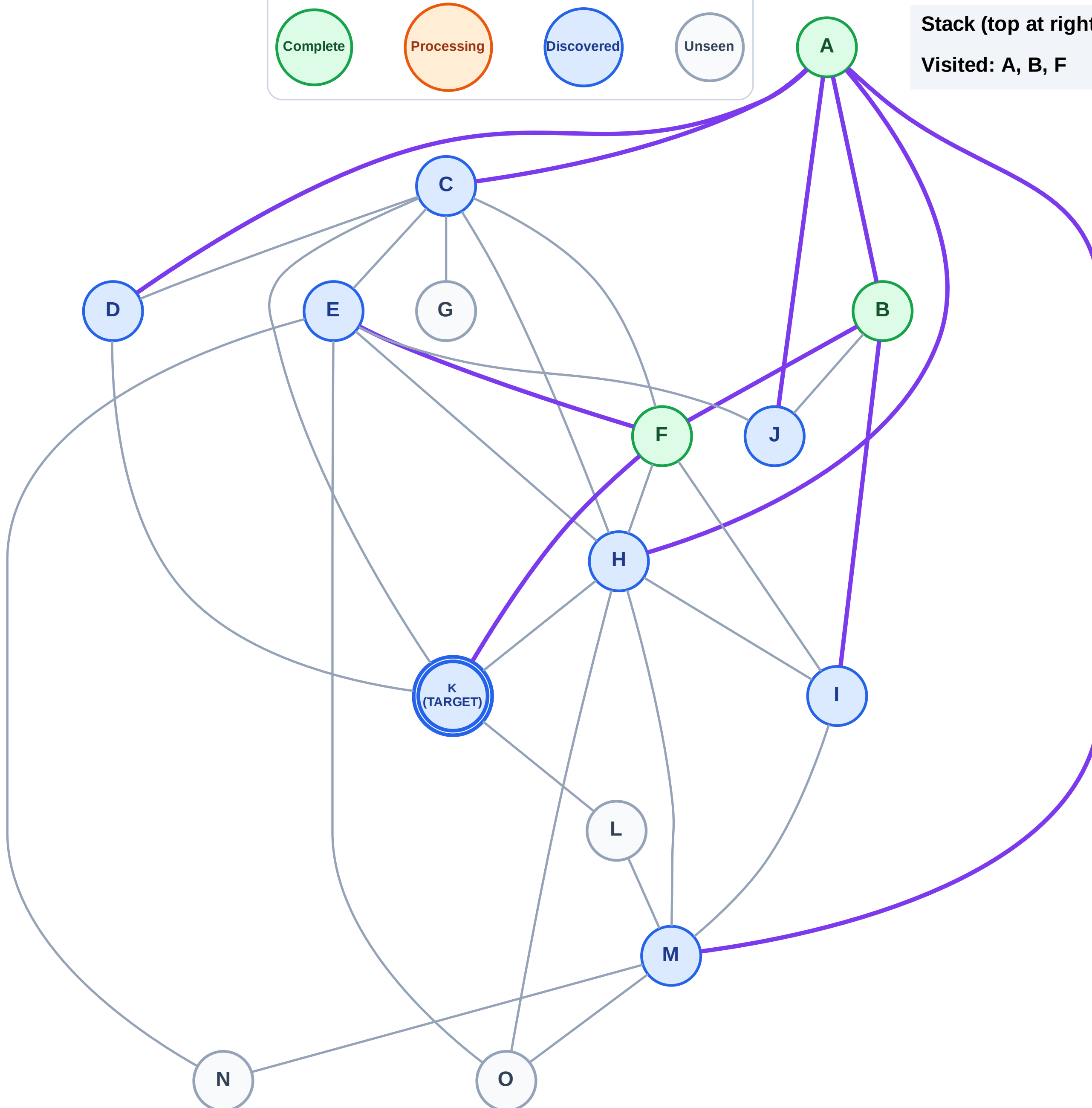


Step 7: Discover K, E from F

Legend

Unseen

Visited: A, B, F



DFS Traversal

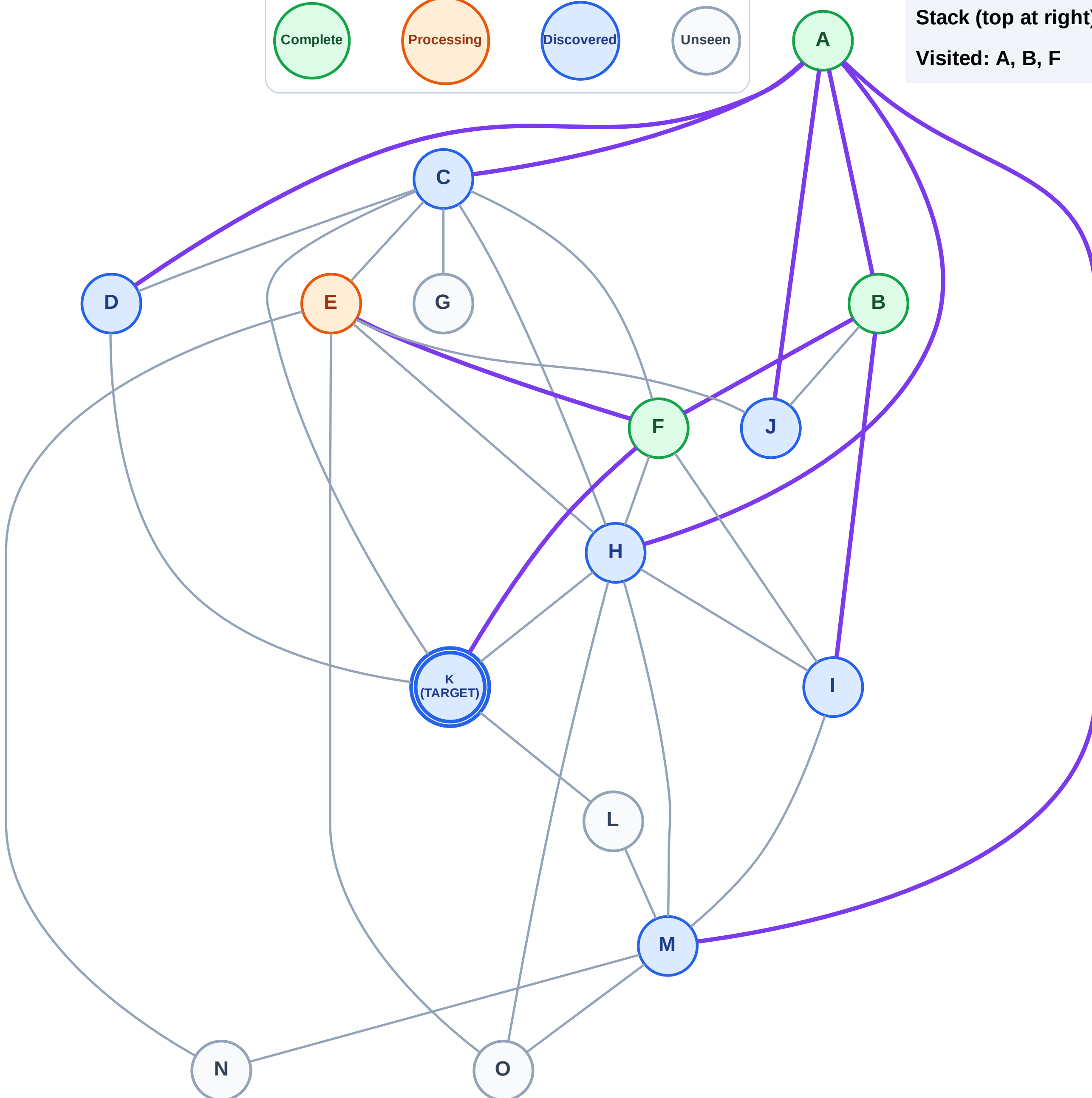
Step 8: Process node E

Legend



Stack (top at right): M | J | H | D | C | I | K

Visited: A, B, F



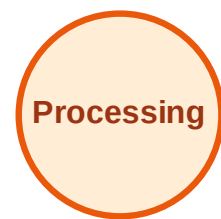
DFS Traversal

Step 9: Discover O, N from E

Legend



Complete



Processing



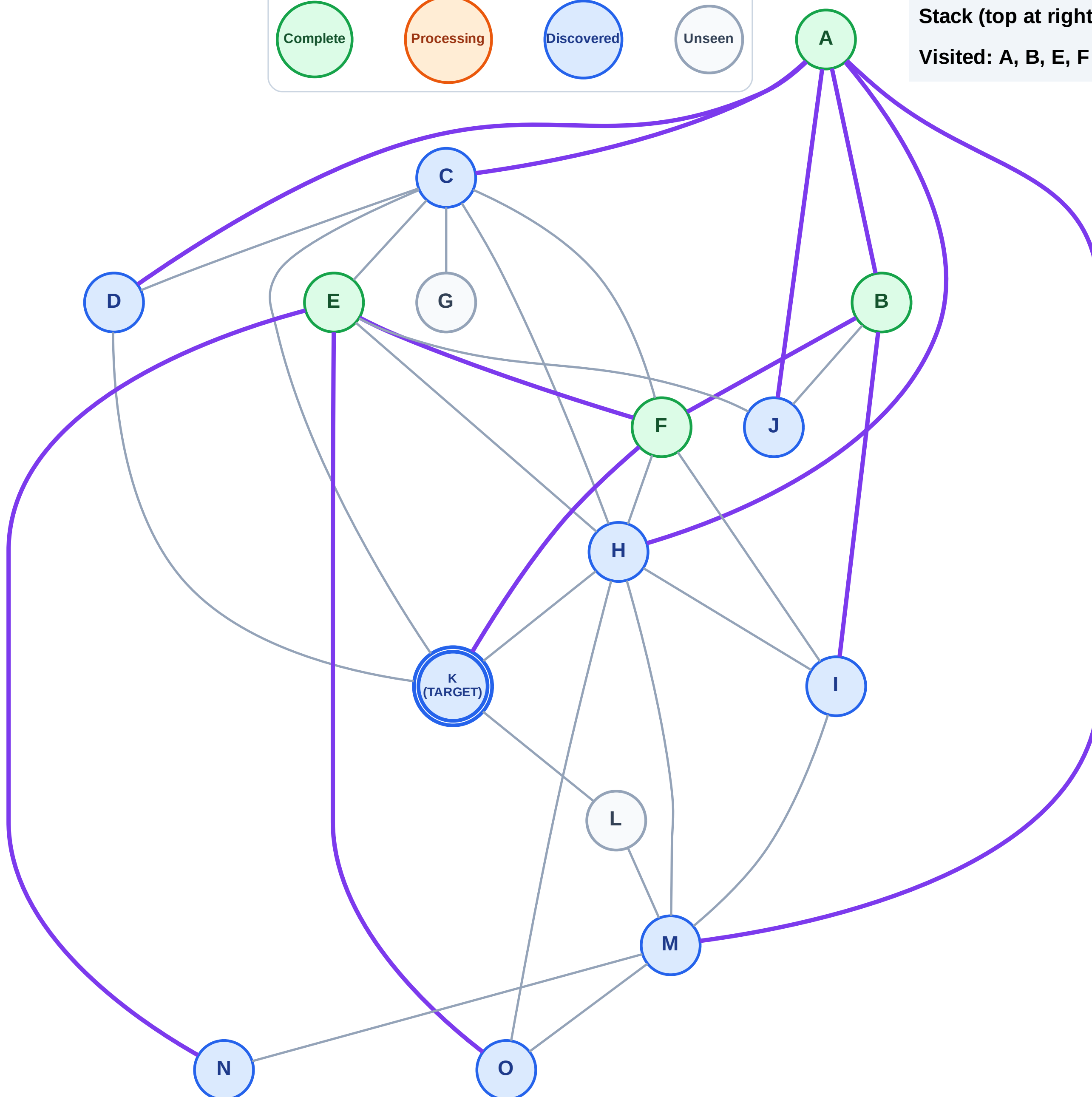
Discovered



Unseen

Stack (top at right): M | J | H | D | C | I | K | O | N

Visited: A, B, E, F



DFS Traversal

Step 10: Process node N

Legend

Complete

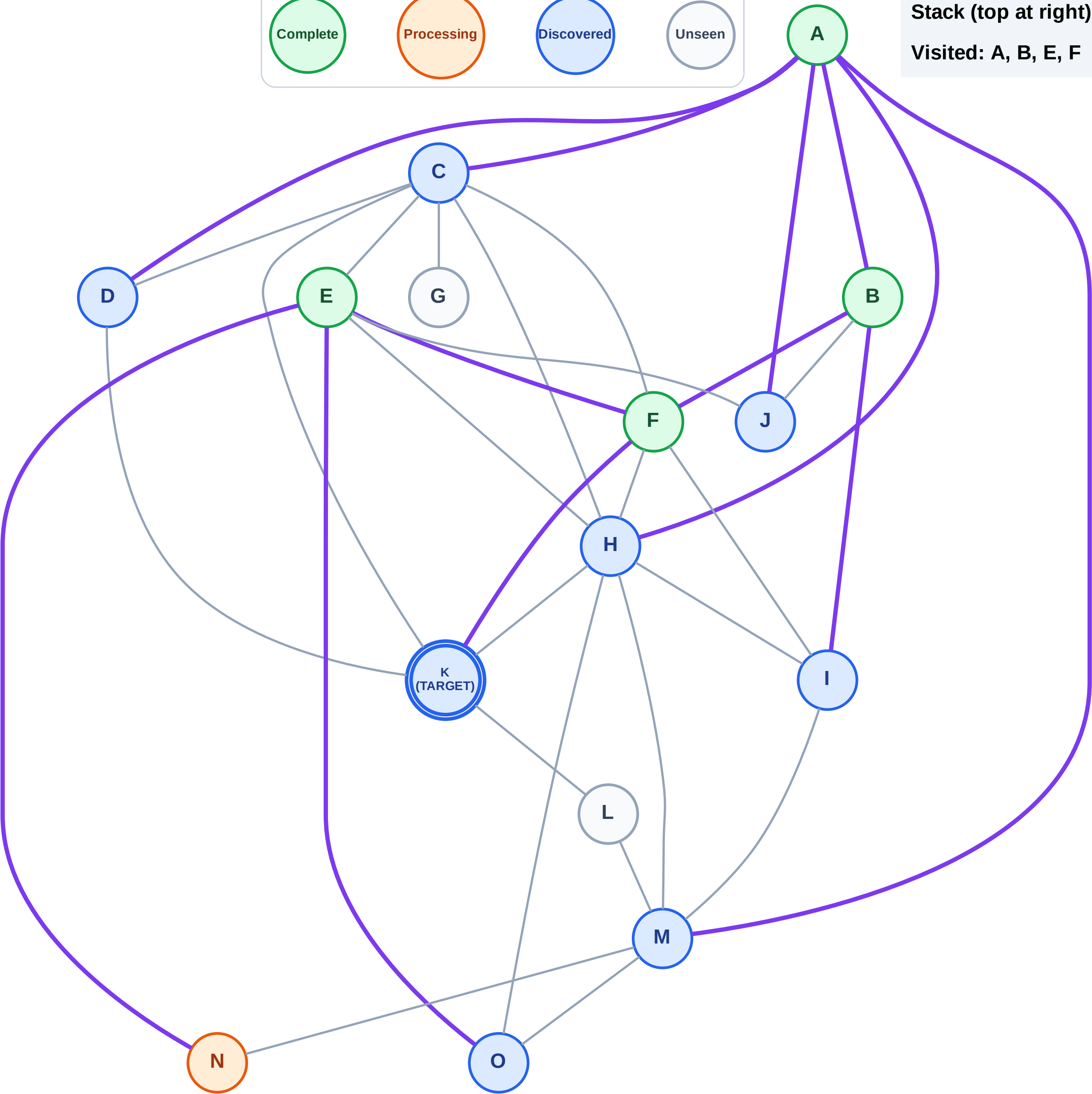
Processing

Discovered

Unseen

Stack (top at right): M | J | H | D | C | I | K | O

Visited: A, B, E, F



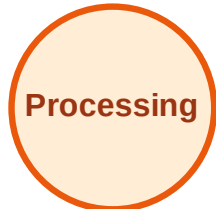
DFS Traversal

Step 11: No new neighbors from N

Legend



Complete



Processing



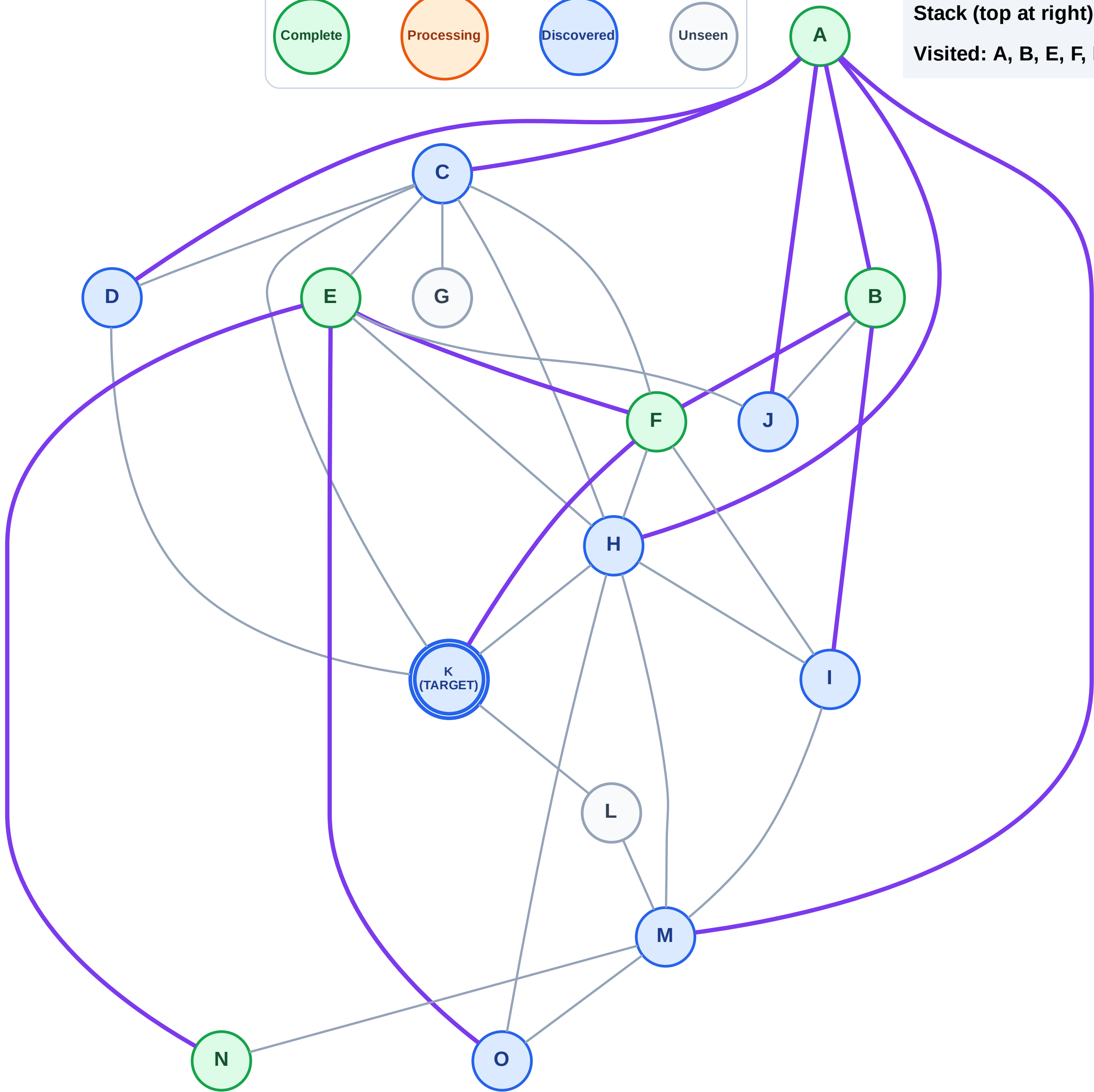
Discovered



Unseen

Stack (top at right): M | J | H | D | C | I | K | O

Visited: A, B, E, F, N



DFS Traversal

Step 12: Process node O

Legend

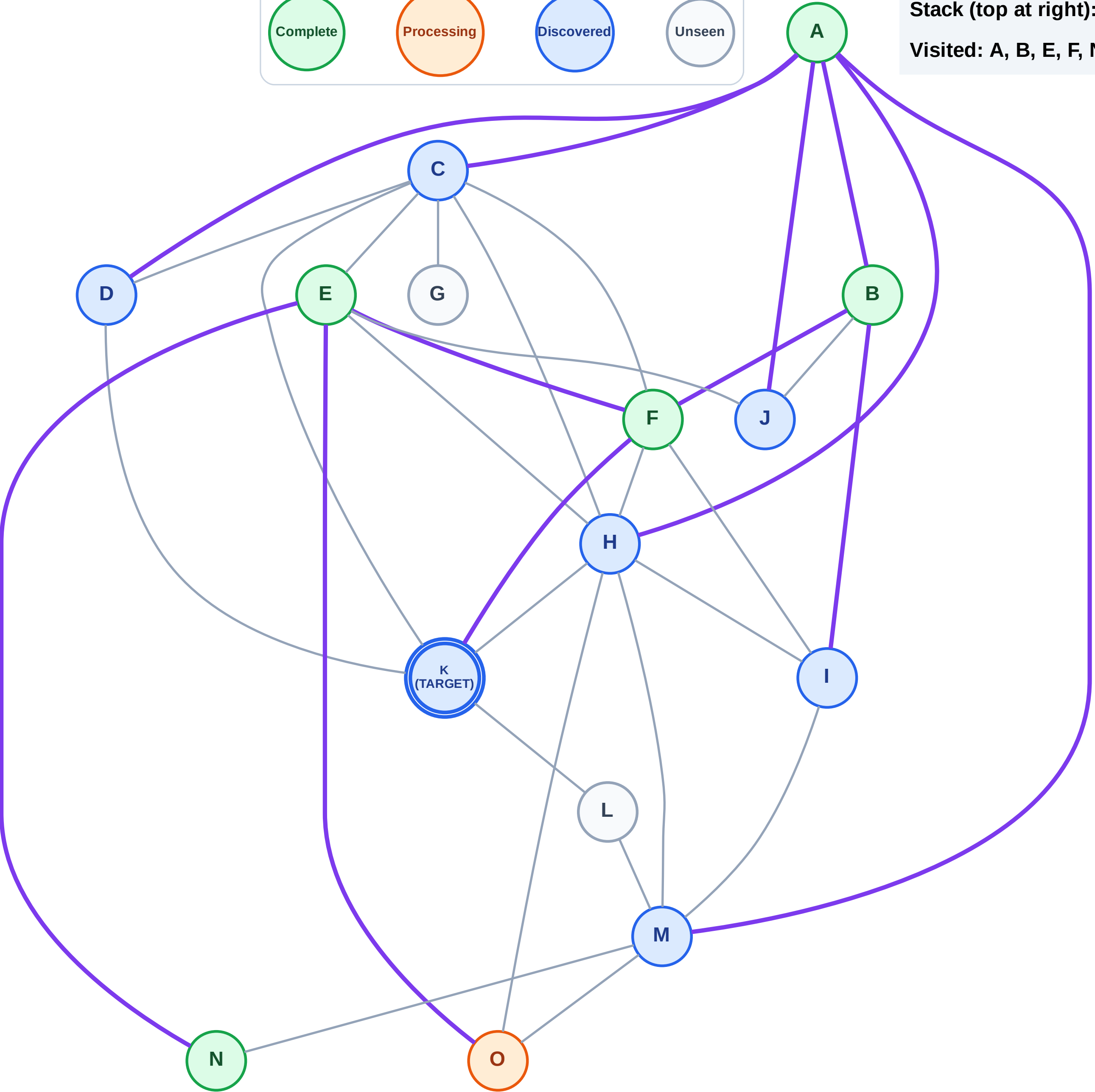
Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | H | D | C | I | K
Visited: A, B, E, F, N



DFS Traversal

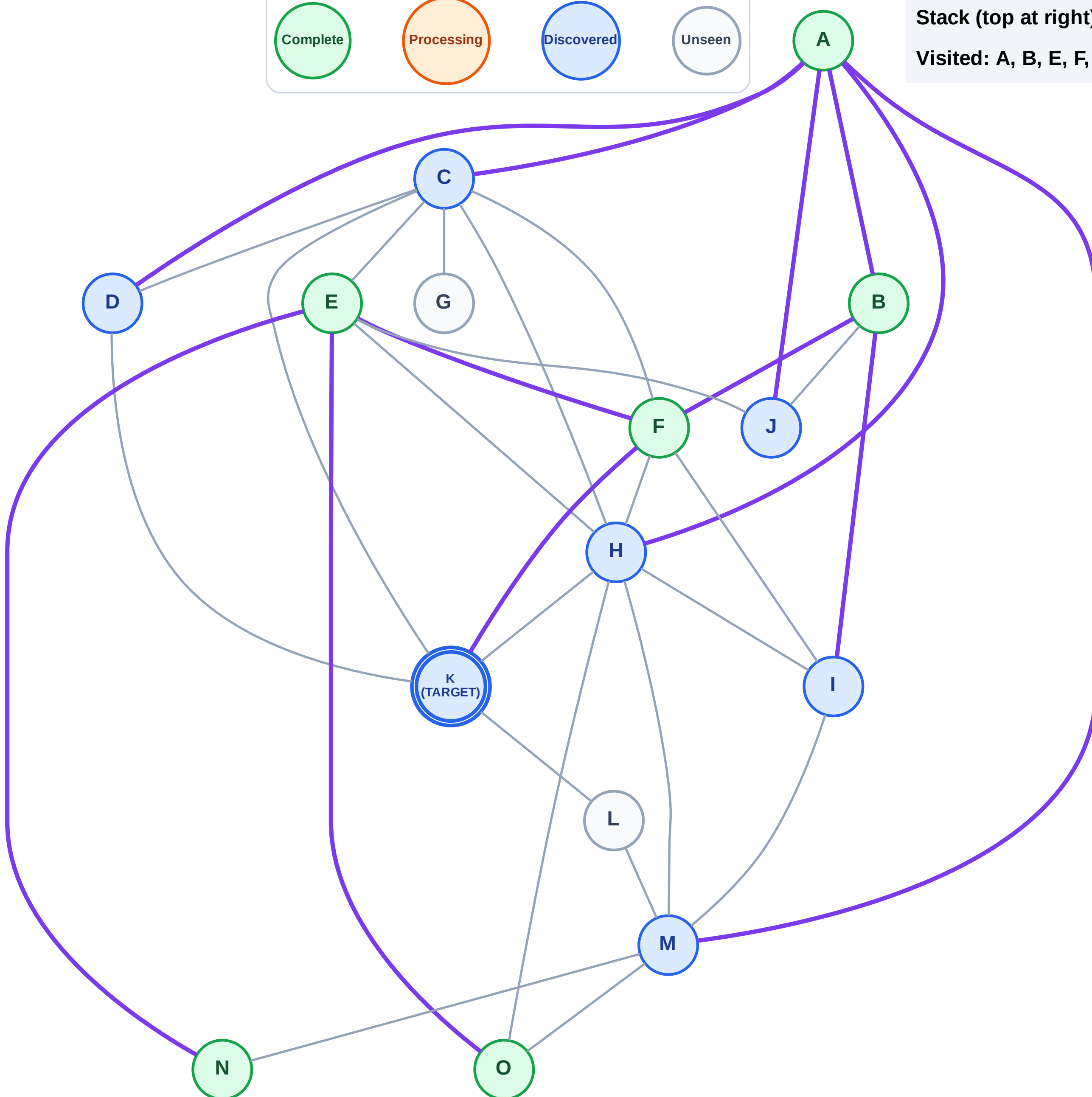
Step 13: No new neighbors from O

Legend



Stack (top at right): M | J | H | D | C | I | K

Visited: A, B, E, F, N, O



DFS Traversal

Step 14: Process node K

Legend

Complete

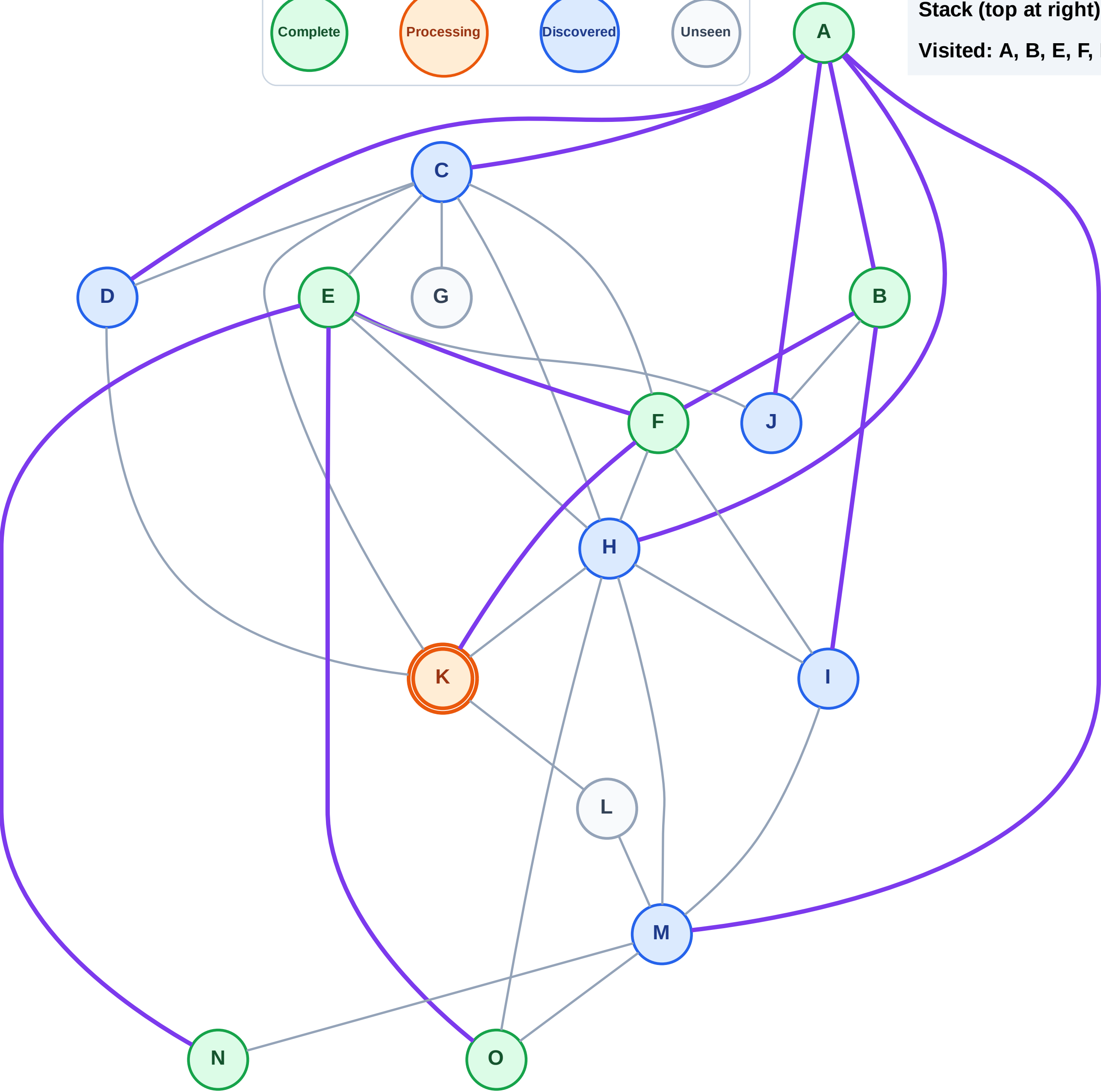
Processing

Discovered

Unseen

Stack (top at right): M | J | H | D | C | I

Visited: A, B, E, F, N, O



DFS Traversal

Step 15: TARGET NODE K REACHED! Path (3 hops): A → B → F → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | H | D | C | I

Visited: A, B, E, F, K, N, O

